

Alligator Harbor Aquatic Preserve

SEACAR Water Quality Analysis

Last compiled on 15 July, 2026

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Indicators

Nutrients

Total Nitrogen - Discrete

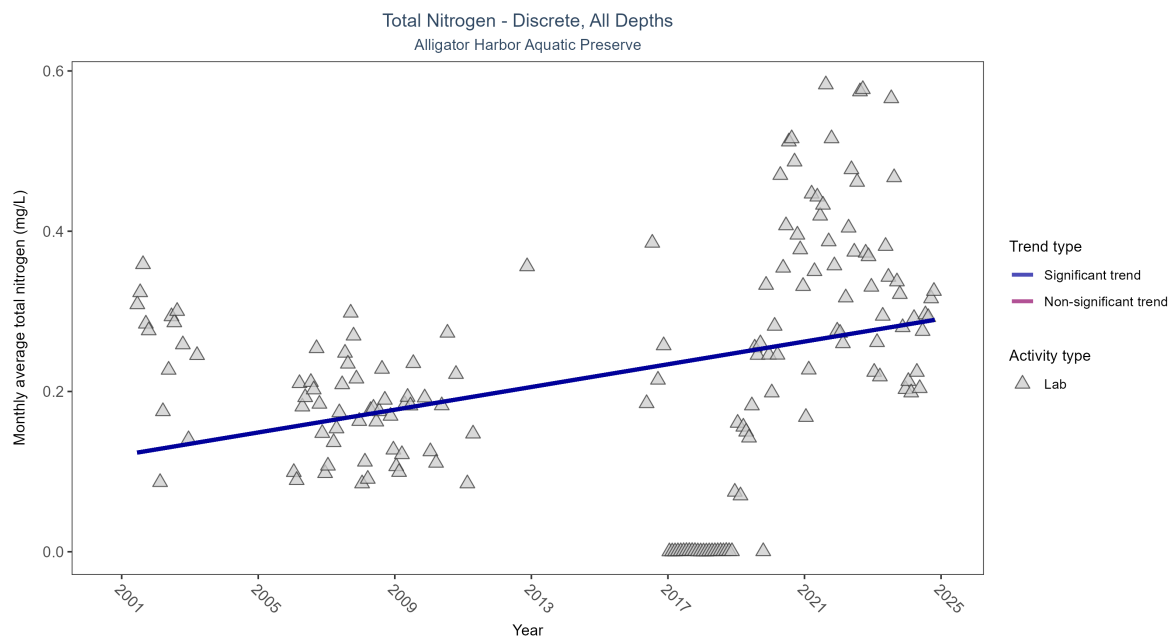


Figure 1: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 1: Monthly average total nitrogen increased by 0.01 mg/L per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	1625	19	2001 - 2024	0.2	0.27903	0.12051	0.00709	0

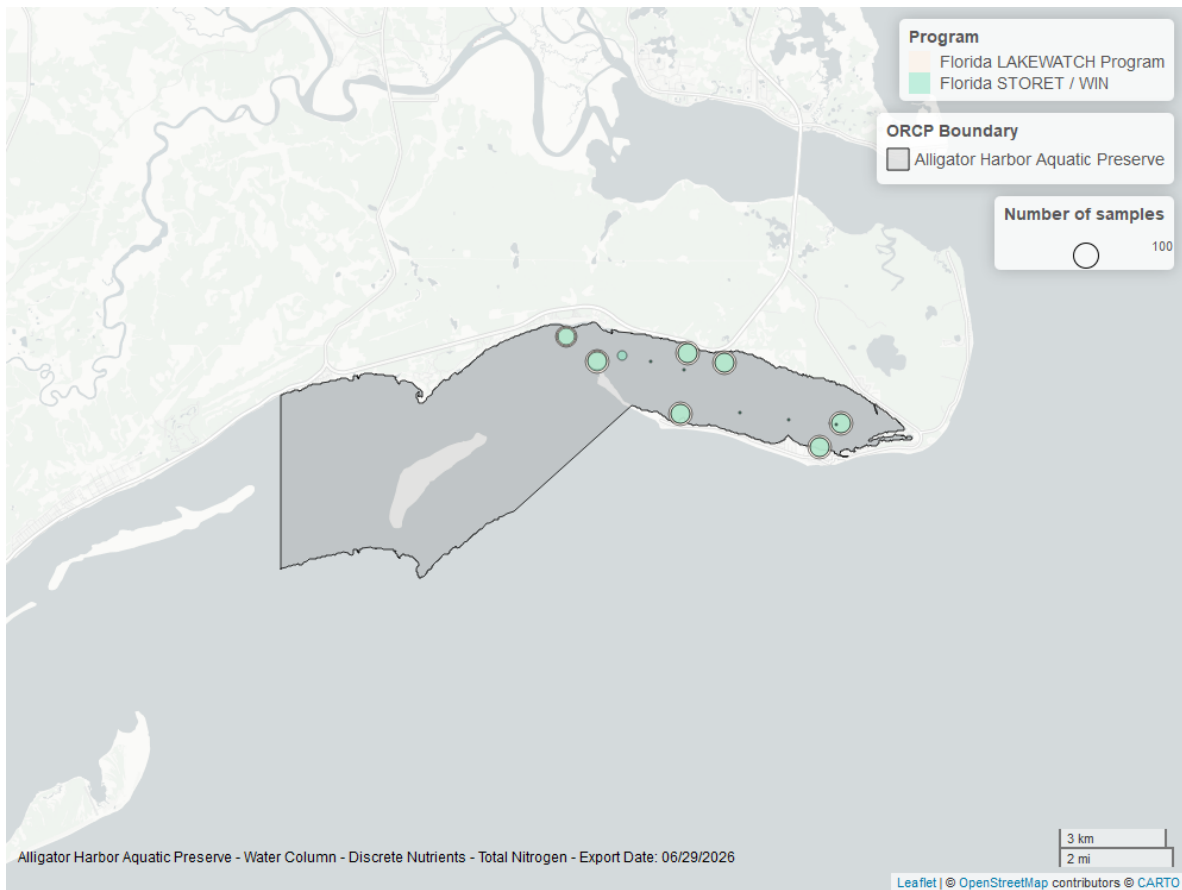


Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Alligator Harbor Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Phosphorus - Discrete

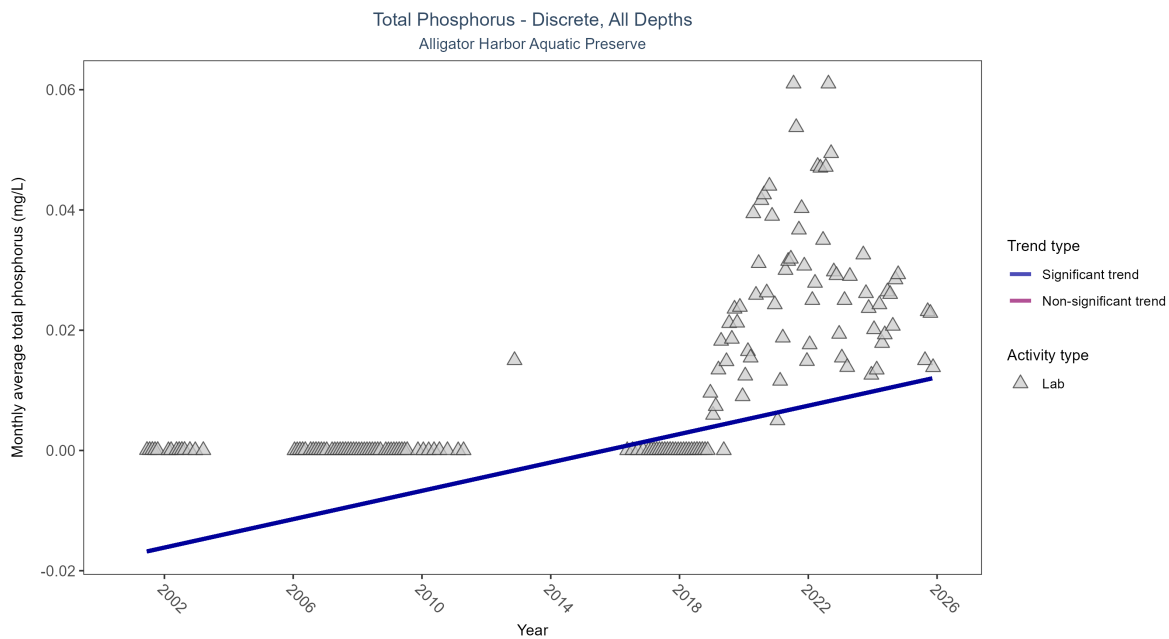


Figure 3: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 2: Monthly average total phosphorus increased by less than 0.01 mg/L per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	1157	20	2001 - 2025	0.00005	0.53779	-0.01733	0.00118	0

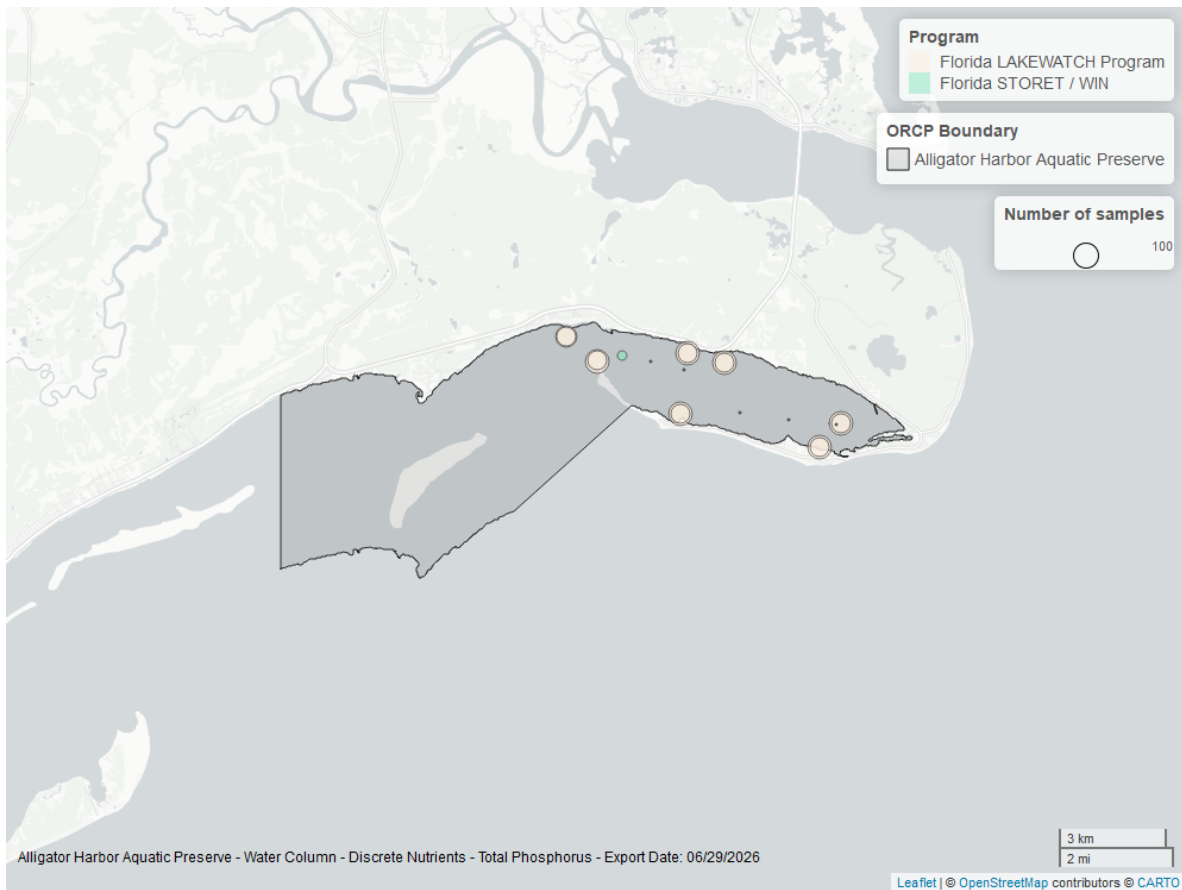


Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Alligator Harbor Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Quality

Dissolved Oxygen - Continuous

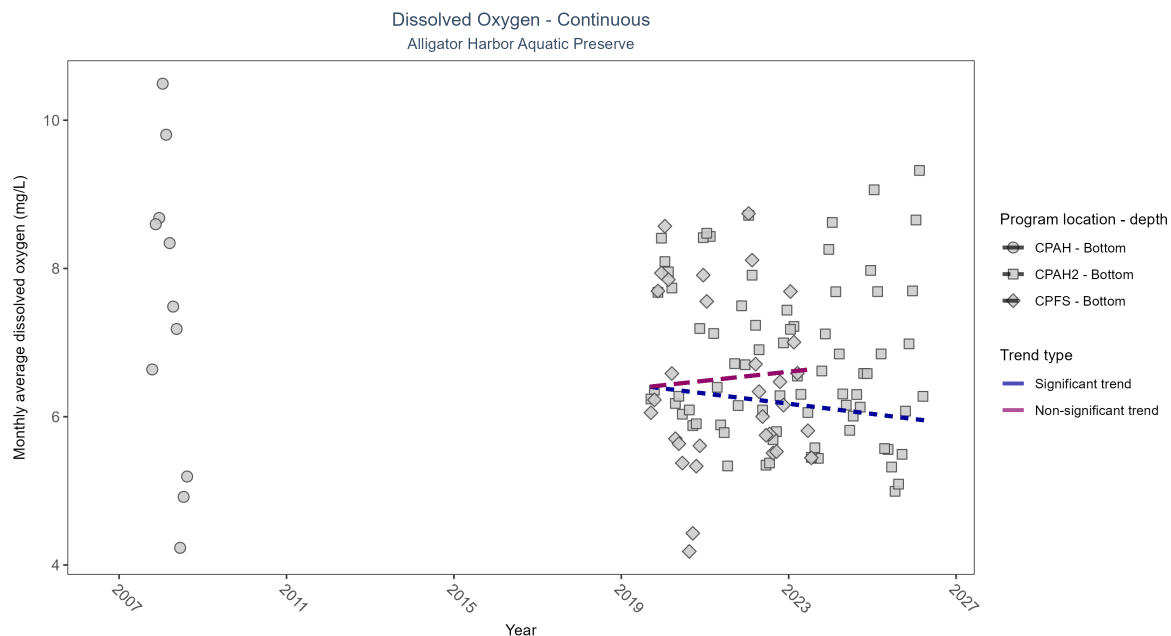


Figure 5: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 3: At one program location, monthly average dissolved oxygen decreased by 0.07 mg/L per year. No detectable change in monthly average dissolved oxygen was observed at one location. There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
CPAH	Insufficient data	11557	2	2007 - 2008	8.0	-	-	-	-
CPAH2	Decreasing trend	187905	8	2019 - 2026	6.7	-0.22	6.45	-0.07	0.03
CPFS	No detectable trend	56853	5	2019 - 2023	6.4	0.12	6.36	0.06	0.74

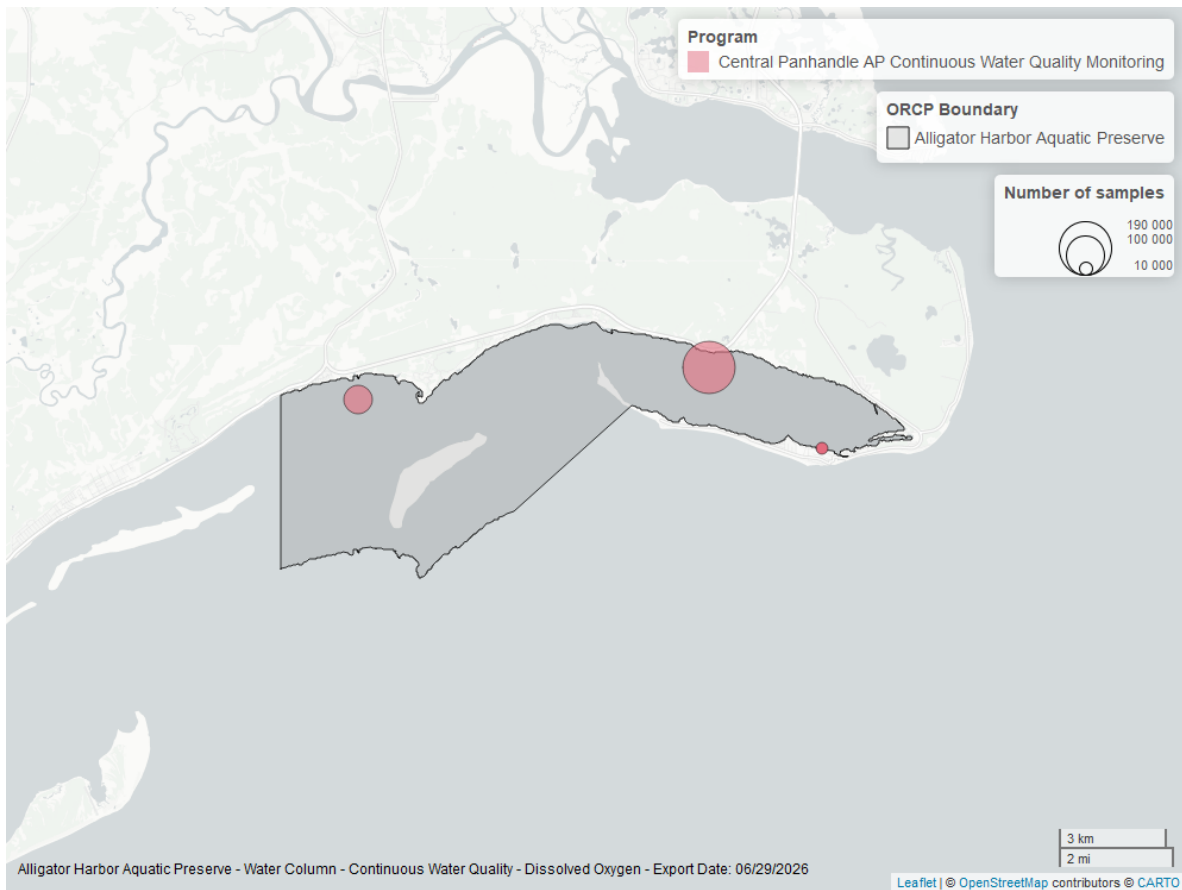


Figure 6: Map showing location of dissolved oxygen continuous water quality sampling locations within the boundaries of *Alligator Harbor Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen - Discrete

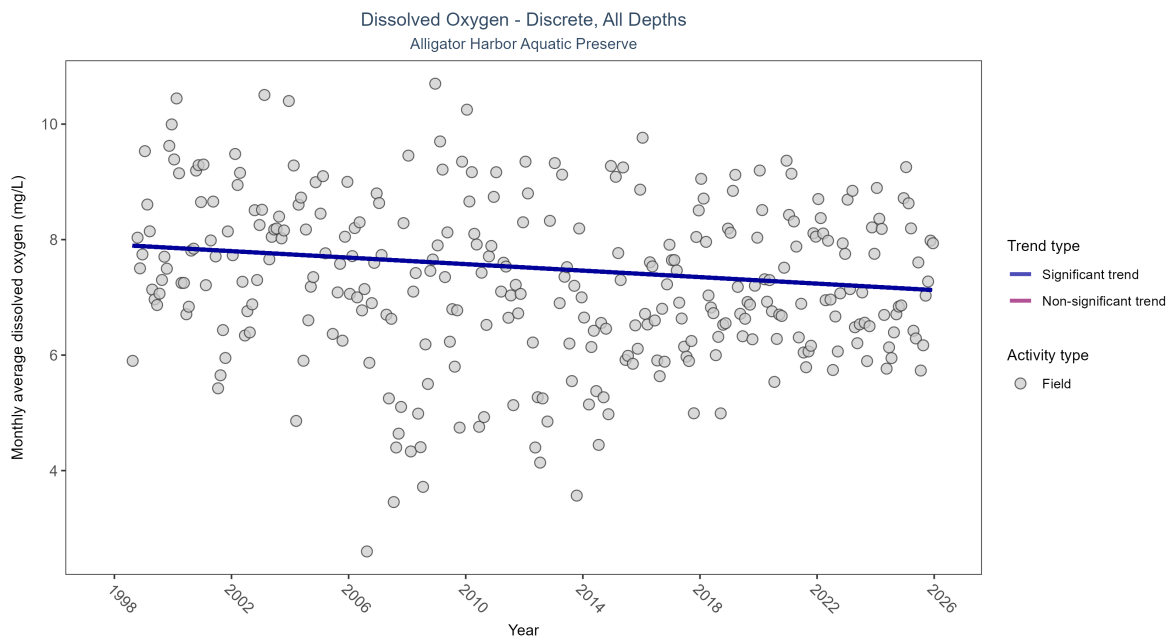


Figure 7: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 4: Monthly average dissolved oxygen decreased by 0.03 mg/L per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	8457	28	1998 - 2025	7.2	-0.16231	7.91292	-0.02812	0.0001

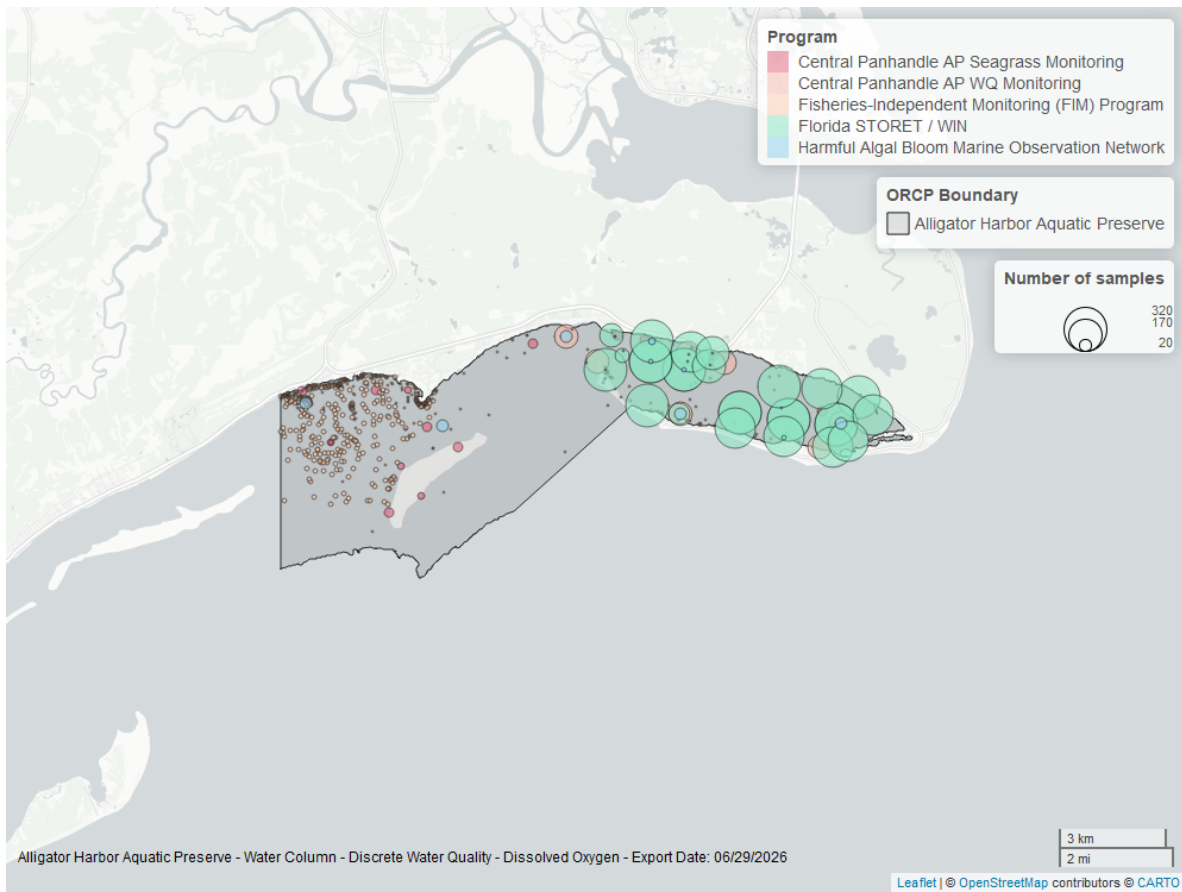


Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Alligator Harbor Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Continuous

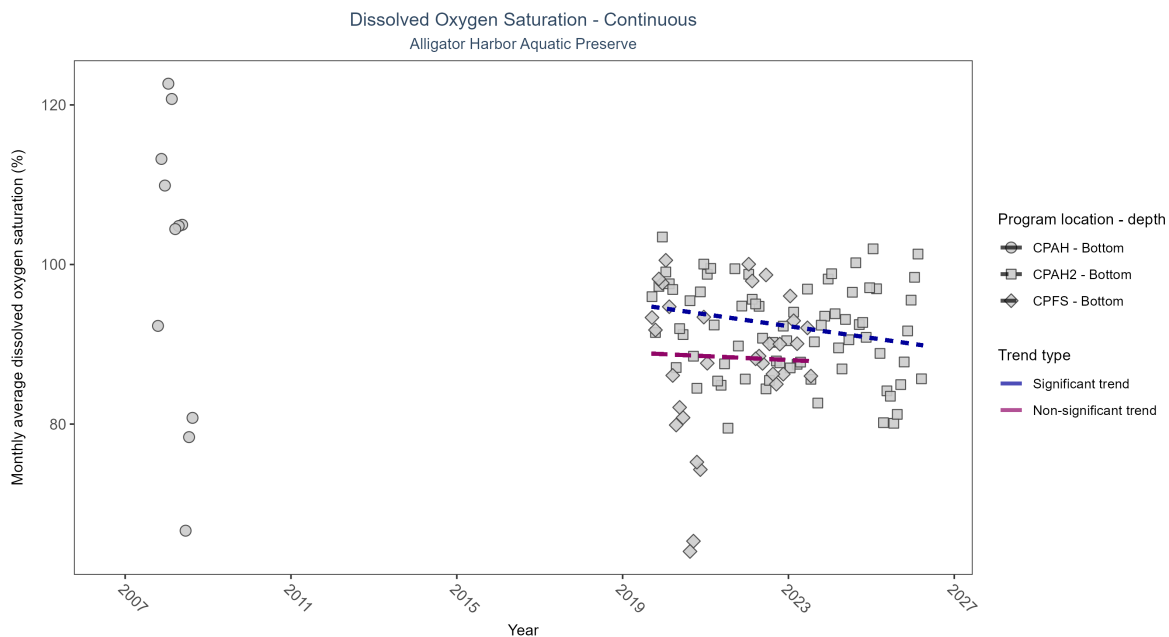


Figure 9: Scatter plot of monthly average dissolved oxygen saturation over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

\begin{table}[H] \caption{At one program location, monthly average dissolved oxygen saturation decreased by 0.74% per year. No detectable change in monthly average dissolved oxygen saturation was observed at one location. There was insufficient data to fit a model for one location.}

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
CPAH	Insufficient data	11557	2	2007 - 2008	105.6	-	-	-	-
CPAH2	Decreasing trend	187906	8	2019 - 2026	93.1	-0.25	95.24	-0.74	0.01
CPFS	No detectable trend	56928	5	2019 - 2023	90.7	0.02	89.01	-0.24	1

\end{table}

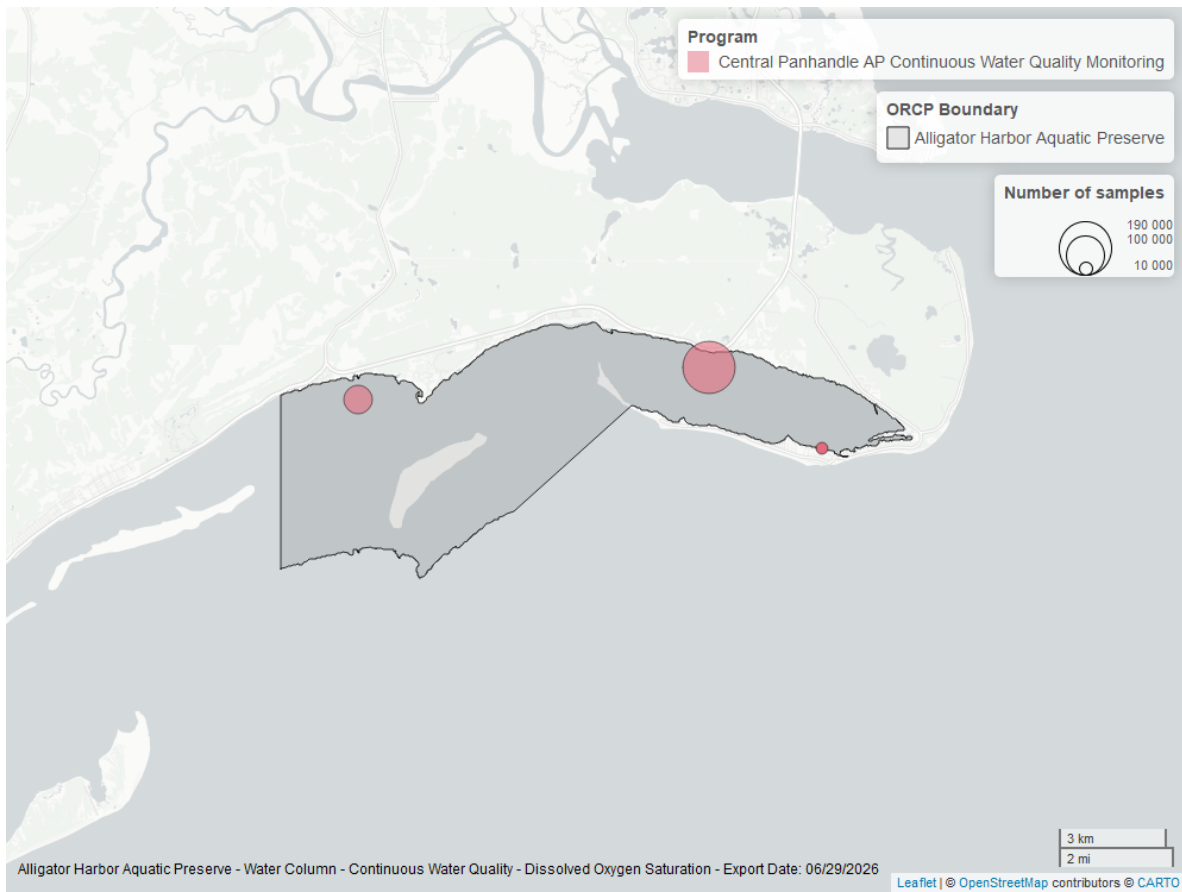


Figure 10: Map showing location of dissolved oxygen saturation continuous water quality sampling locations within the boundaries of *Alligator Harbor Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Discrete

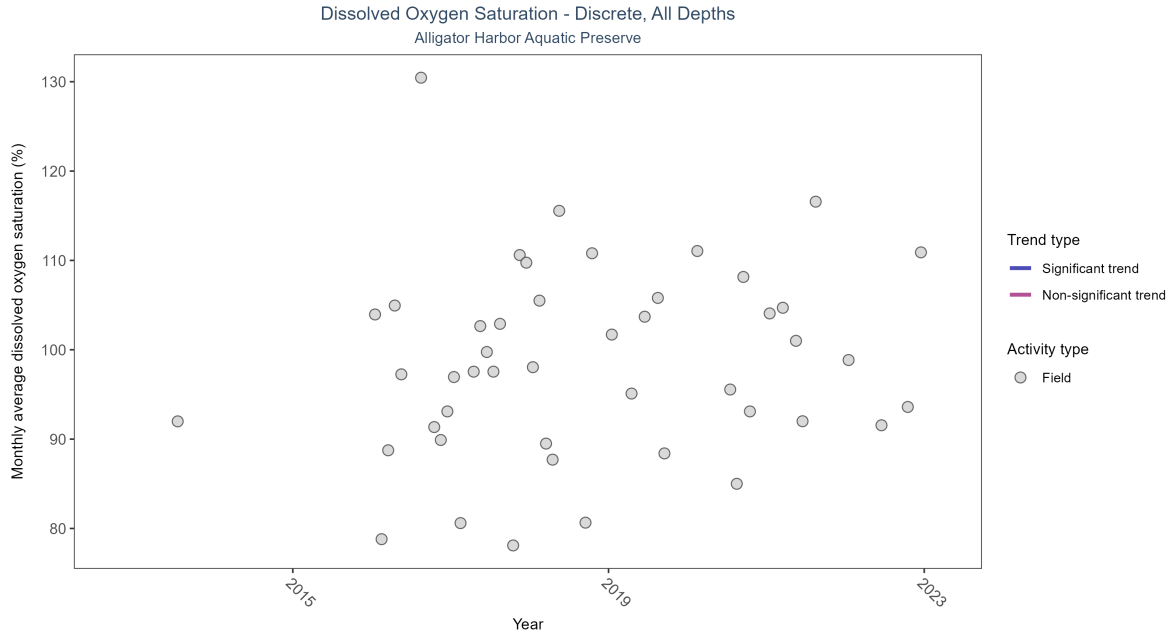


Figure 11: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

Table 5: There was insufficient data to fit a model for dissolved oxygen saturation.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Insufficient data	131	8	2013 - 2022	96.8	-	-	-	-

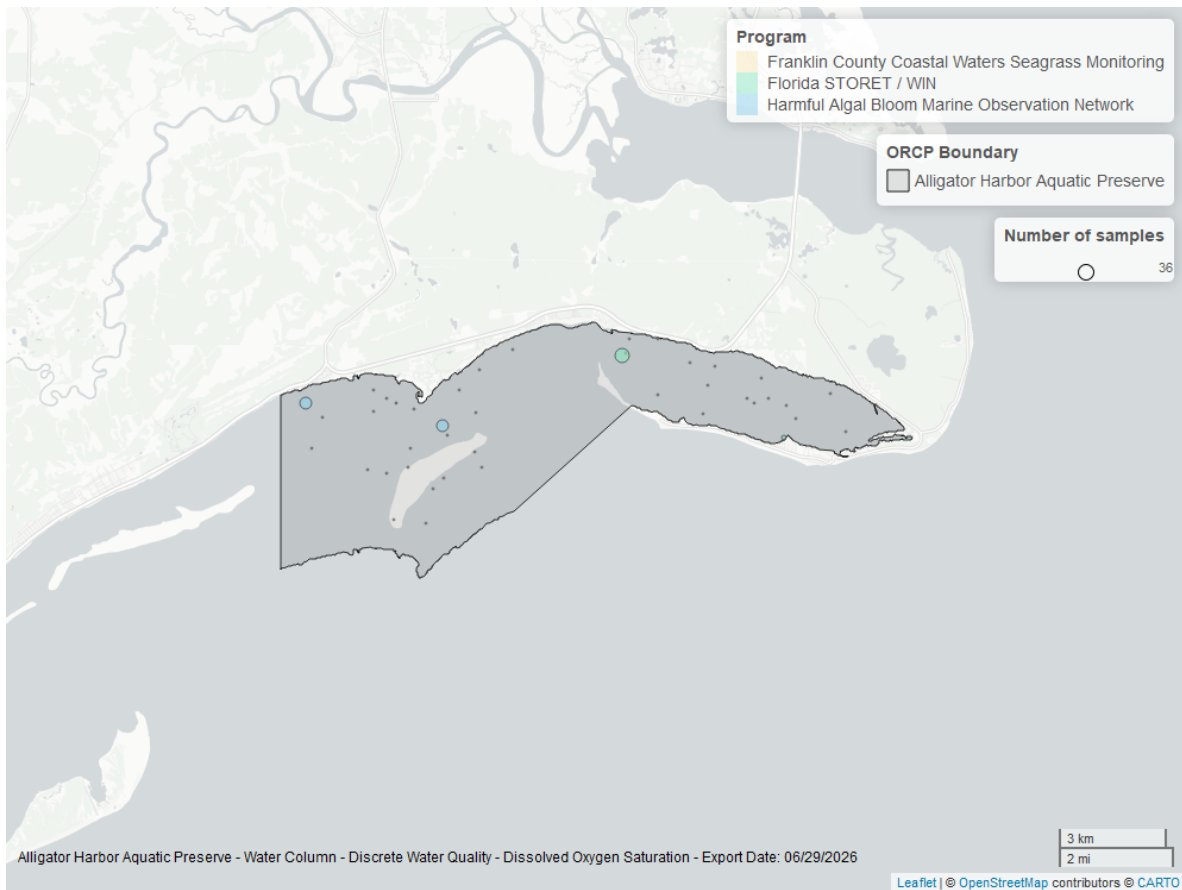


Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Alligator Harbor Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Discrete

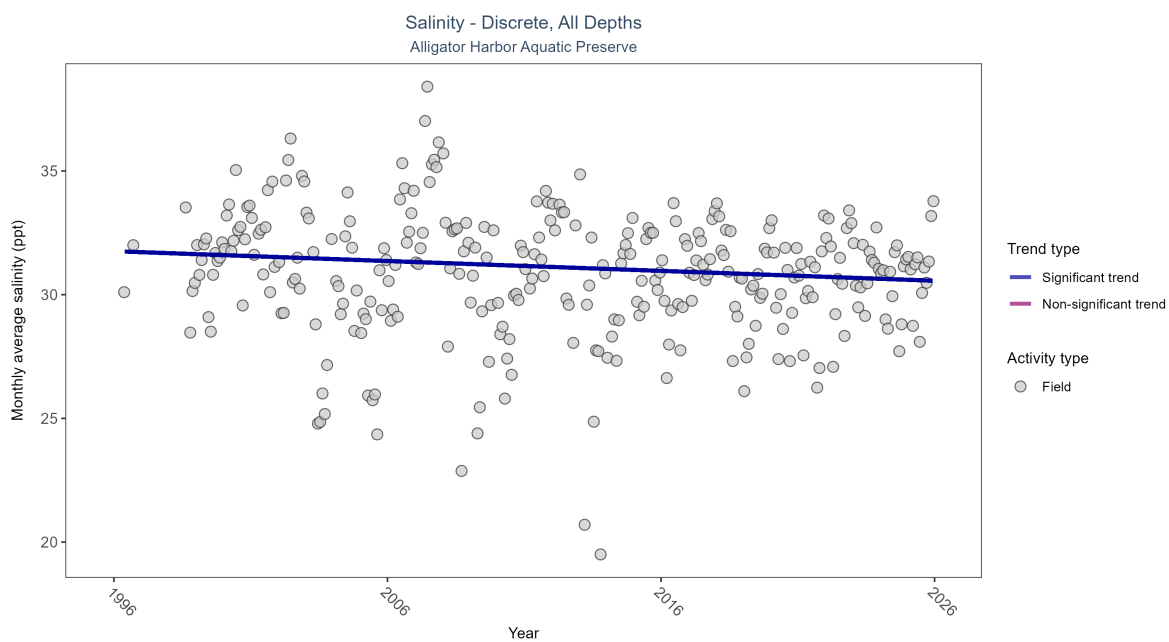


Figure 13: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 6: Monthly average salinity decreased by 0.04 ppt per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	Decreasing trend	9712	29	1996 - 2025	31.1	-0.10895	31.75772	-0.03983	0.0062

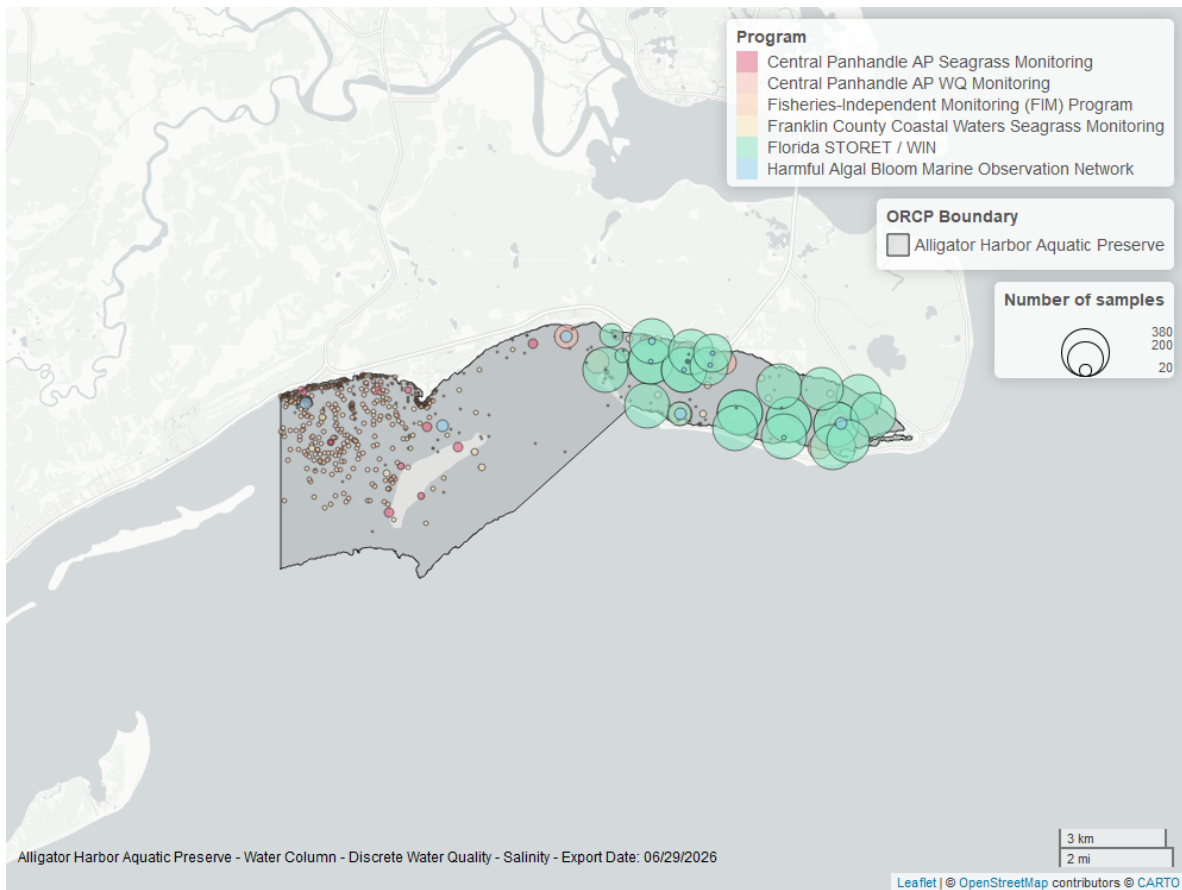


Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Alligator Harbor Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Continuous

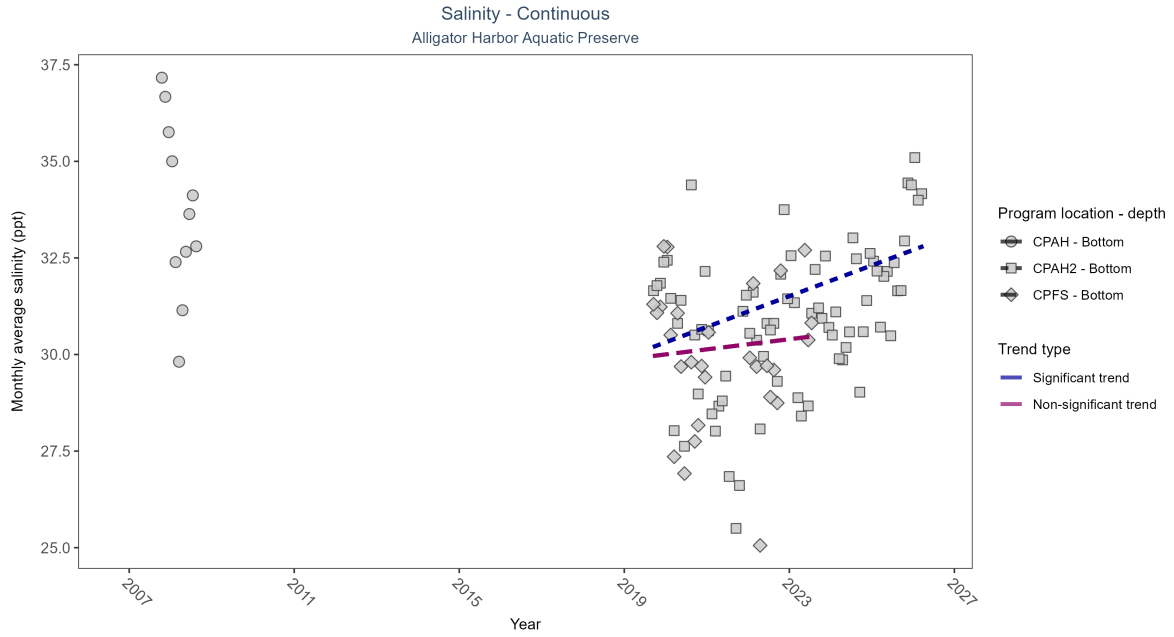


Figure 15: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 7: At one program location, monthly average salinity increased by 0.40 ppt per year. No detectable change in monthly average salinity was observed at one location. There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
CPAH	Insufficient data	13034	2	2007 - 2008	34.0	-	-	-	-
CPAH2	Increasing trend	186148	8	2019 - 2026	31.2	0.3	29.92	0.4	0
CPFS	No detectable trend	60346	5	2019 - 2023	30.5	0	29.87	0.13	1

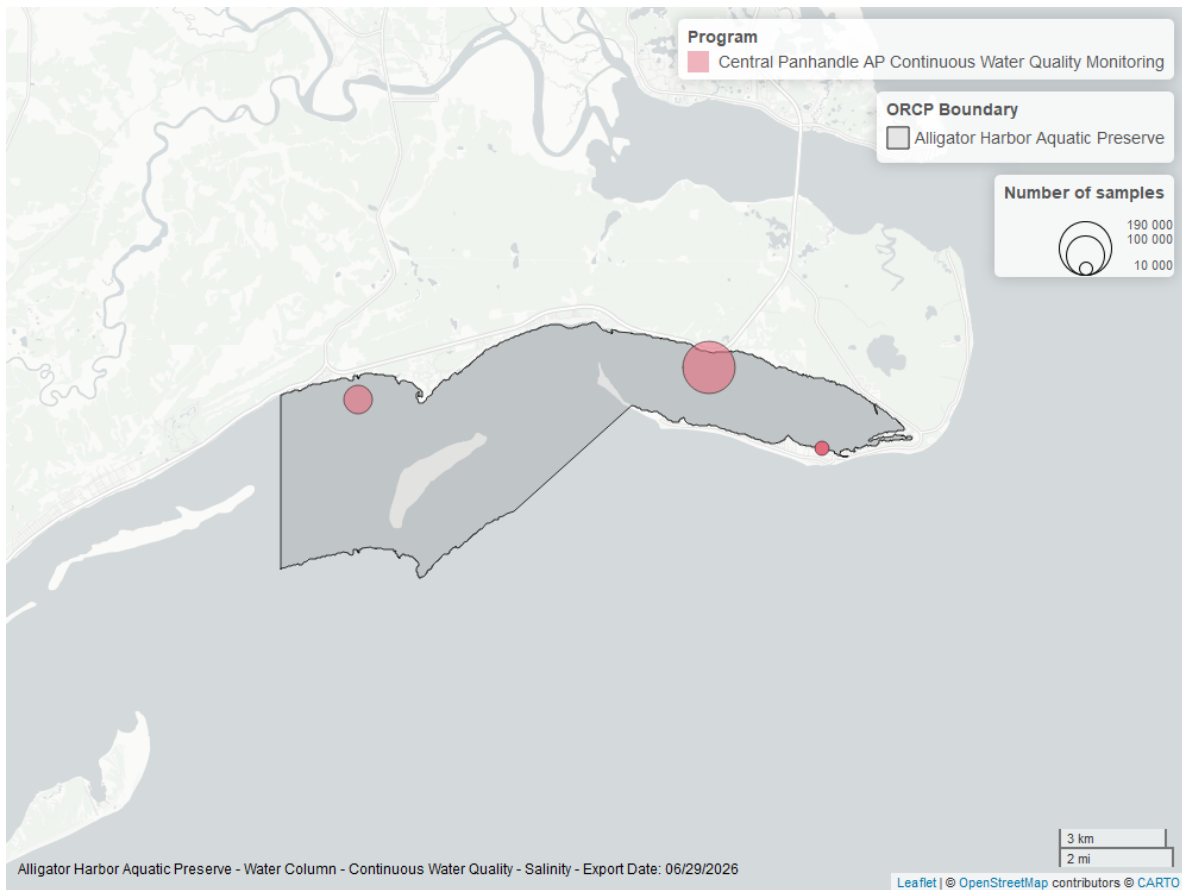


Figure 16: Map showing location of salinity continuous water quality sampling locations within the boundaries of *Alligator Harbor Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Continuous

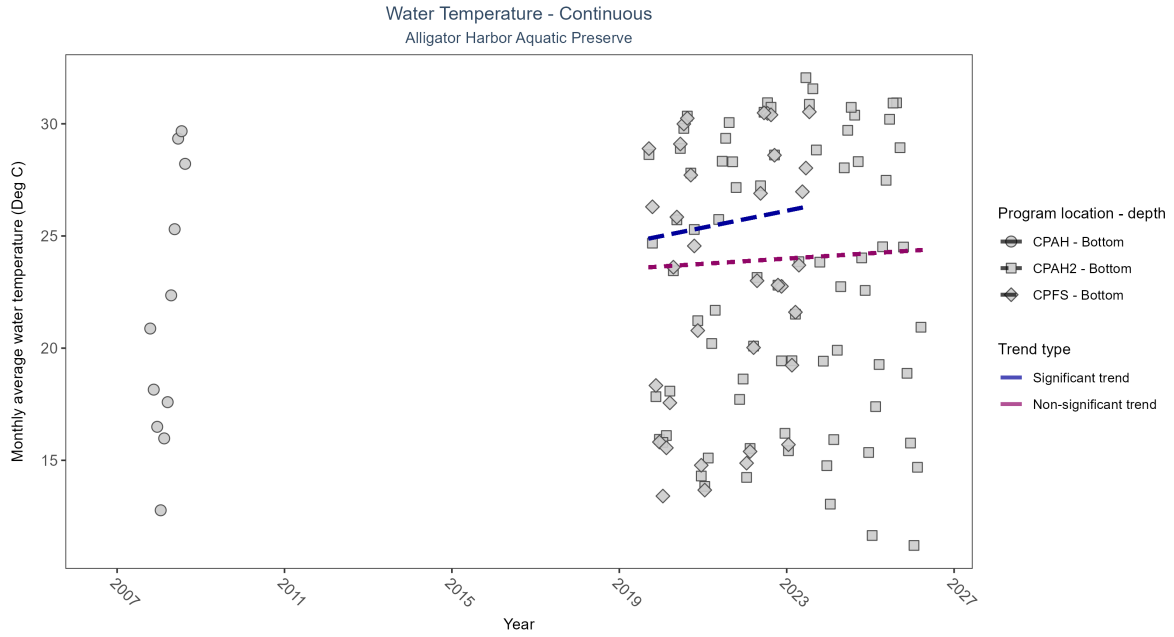


Figure 17: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 8: At one program location, monthly average water temperature increased by 0.38Deg C per year. No detectable change in monthly average water temperature was observed at one location. There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
CPAH	Insufficient data	13034	2	2007 - 2008	20.9	-	-	-	-
CPAH2	No detectable trend	204134	8	2019 - 2026	23.1	0.14	23.52	0.12	0.25
CPFS	Increasing trend	80801	5	2019 - 2023	24.1	0.37	24.62	0.38	0.03

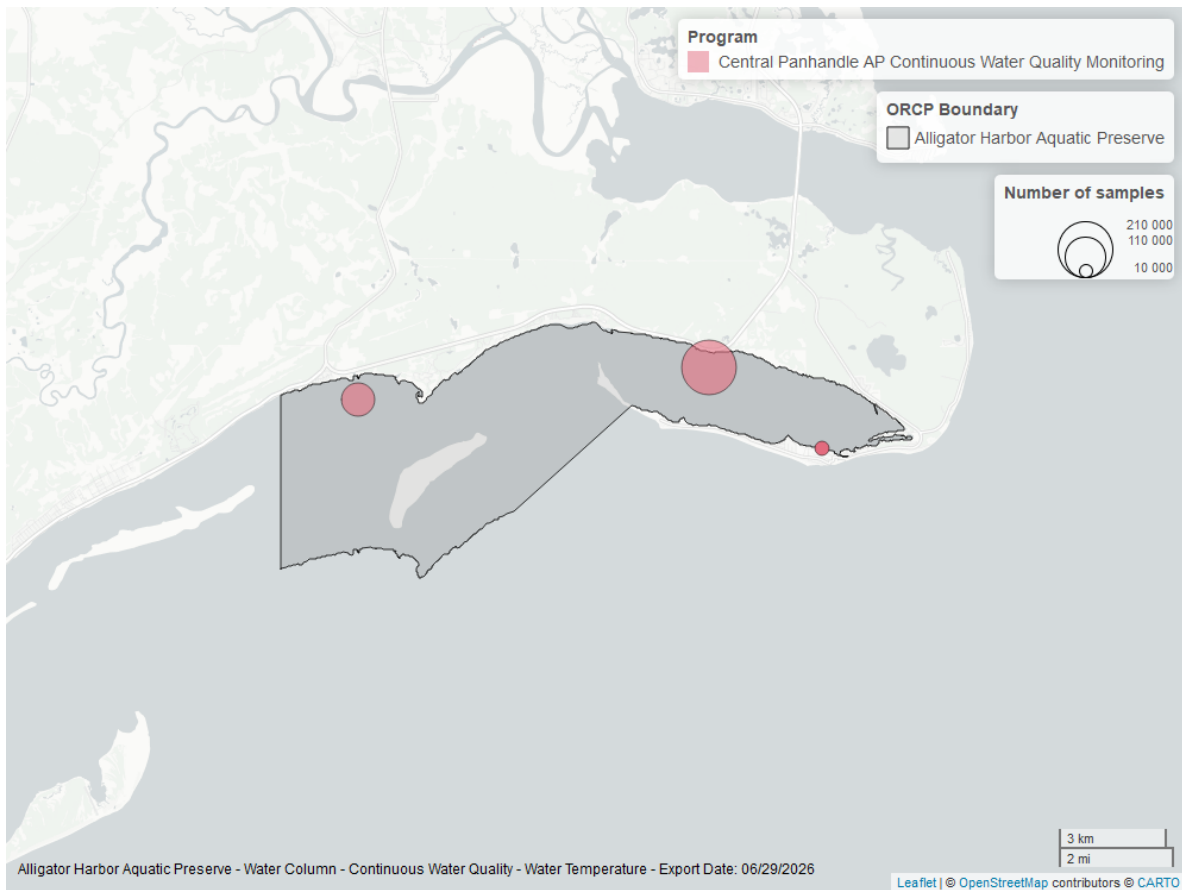


Figure 18: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Alligator Harbor Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Discrete

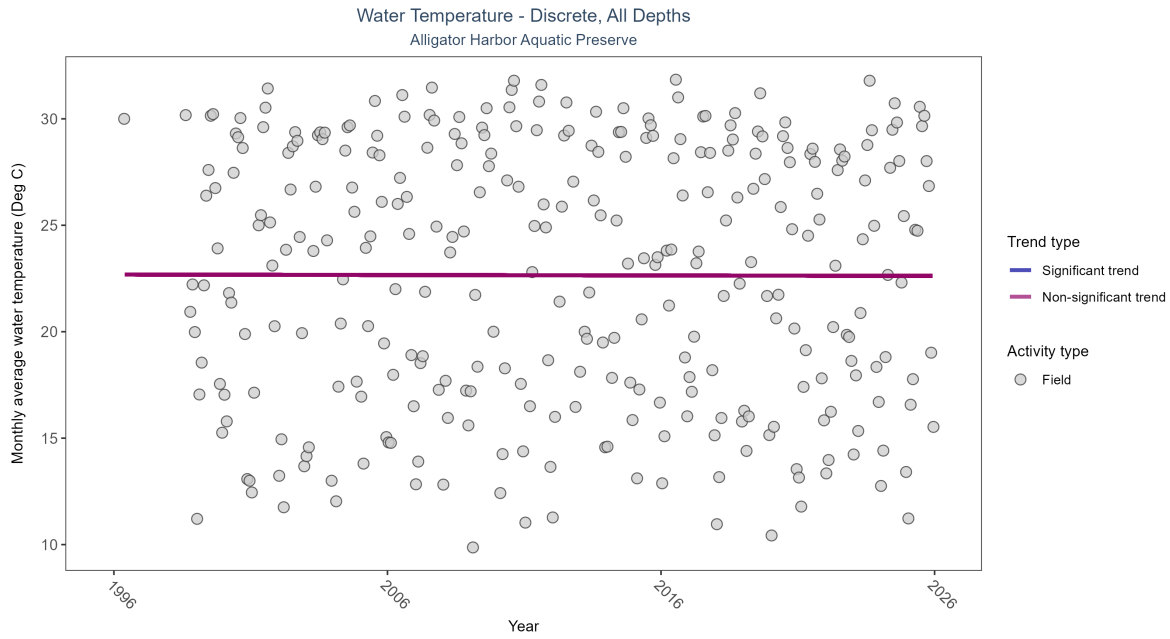


Figure 19: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 9: Water temperature showed no detectable trend between 1996 and 2025.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	9958	29	1996 - 2025	25.9	-0.0056	22.68476	-0.00212	0.8854

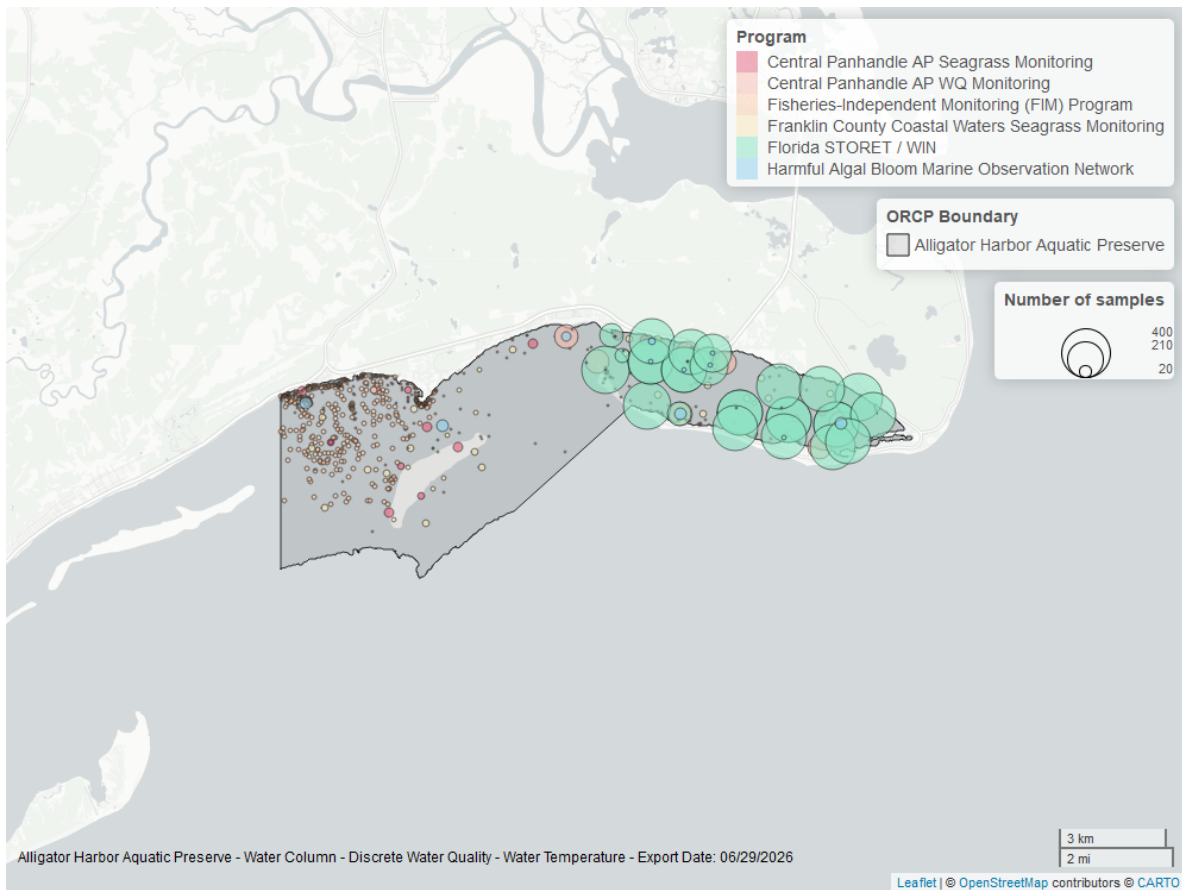


Figure 20: Map showing location of discrete water quality sampling locations within the boundaries of *Alligator Harbor Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

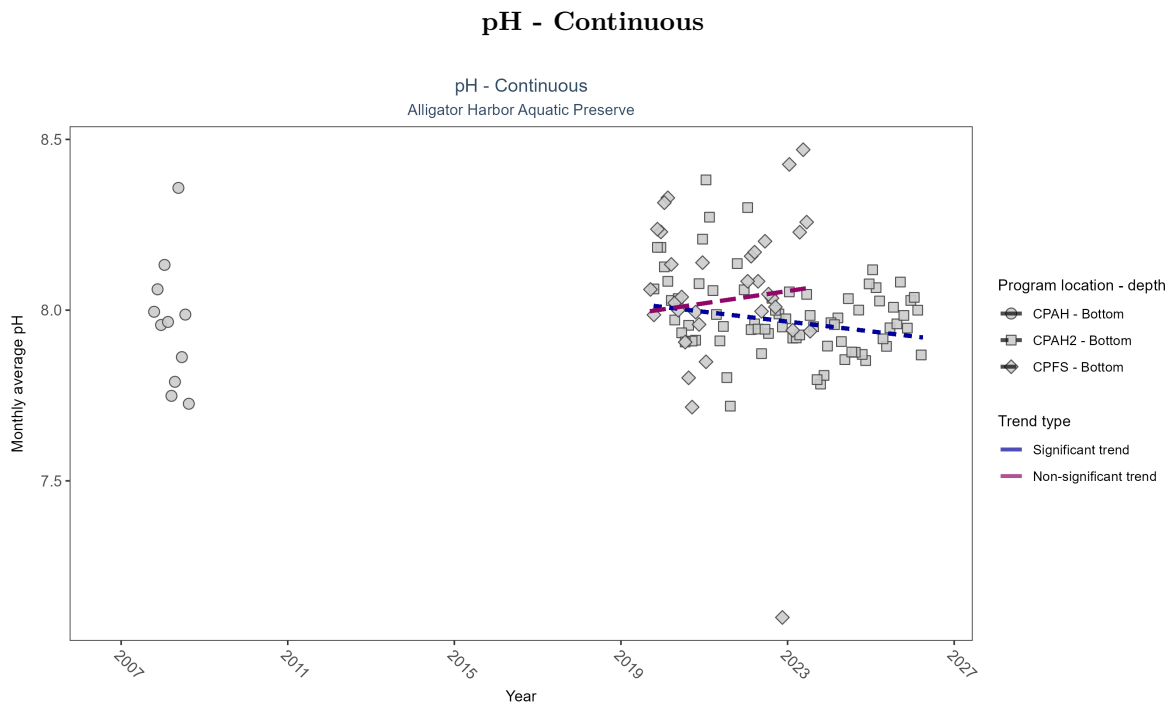


Figure 21: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 10: At one program location, monthly average pH decreased by 0.01 pH units per year. No detectable change in monthly average pH was observed at one location. There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
CPAH	Insufficient data	13034	2	2007 - 2008	7.9	-	-	-	-
CPAH2	Decreasing trend	185428	8	2019 - 2026	8.0	-0.21	8.02	-0.01	0.02
CPFS	No detectable trend	61868	5	2019 - 2023	8.1	0.19	7.98	0.02	0.52

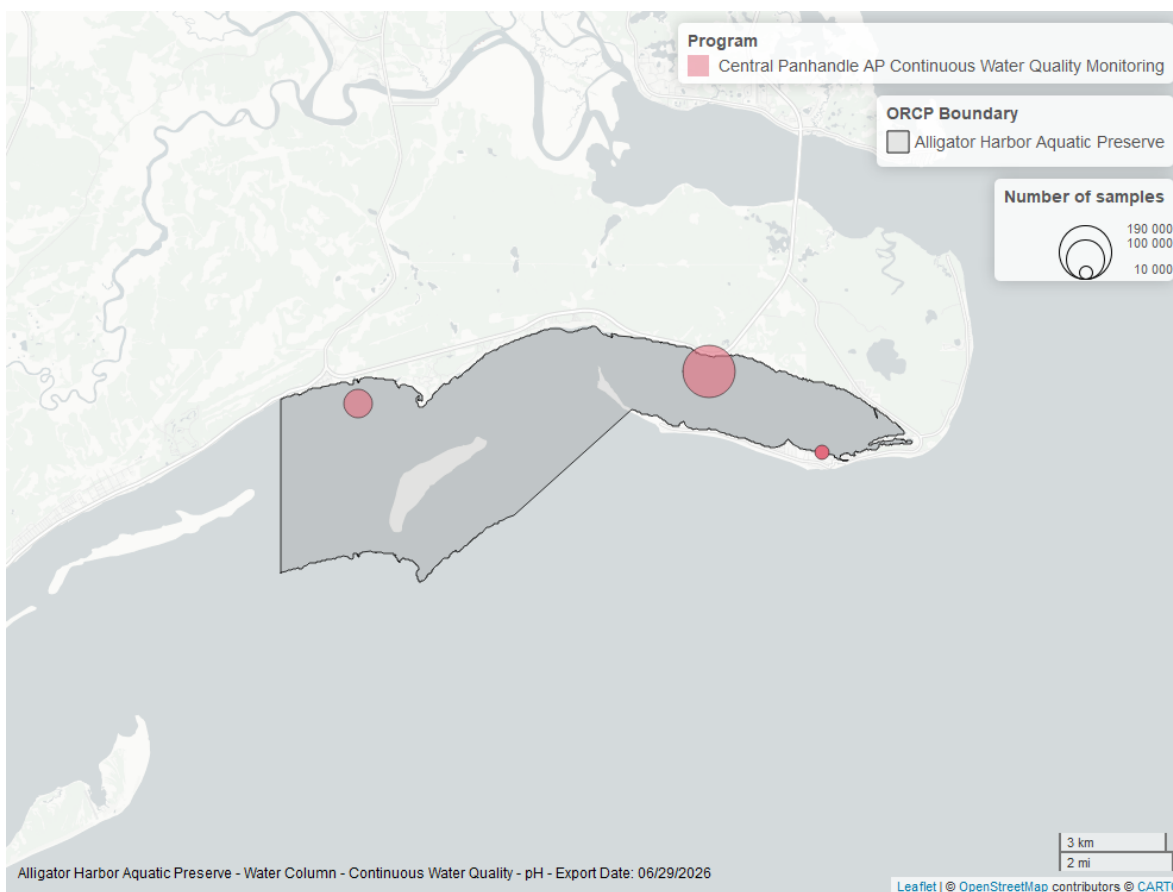


Figure 22: Map showing location of pH continuous water quality sampling locations within the boundaries of *Alligator Harbor Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

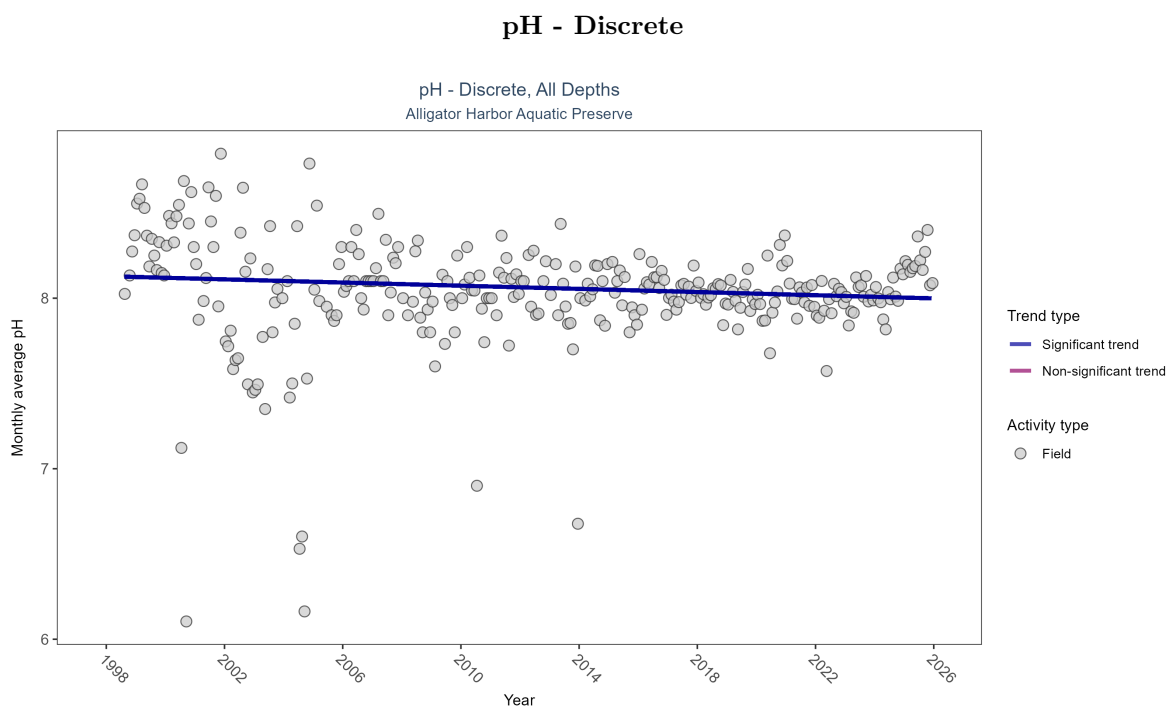


Figure 23: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 11: Monthly average pH decreased by less than 0.01 pH units per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	5564	28	1998 - 2025	8.1	-0.11269	8.129	-0.00465	0.0059

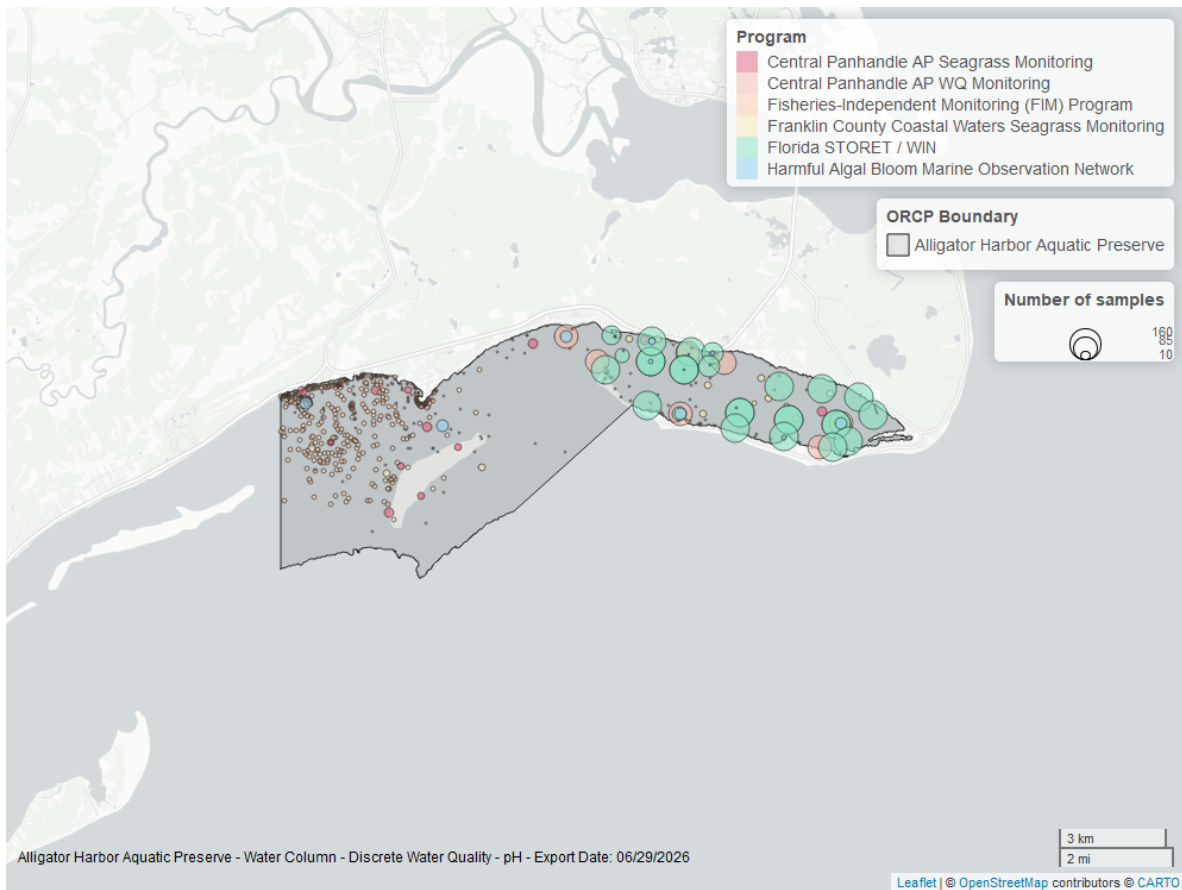


Figure 24: Map showing location of discrete water quality sampling locations within the boundaries of *Alligator Harbor Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Specific Conductivity - Continuous

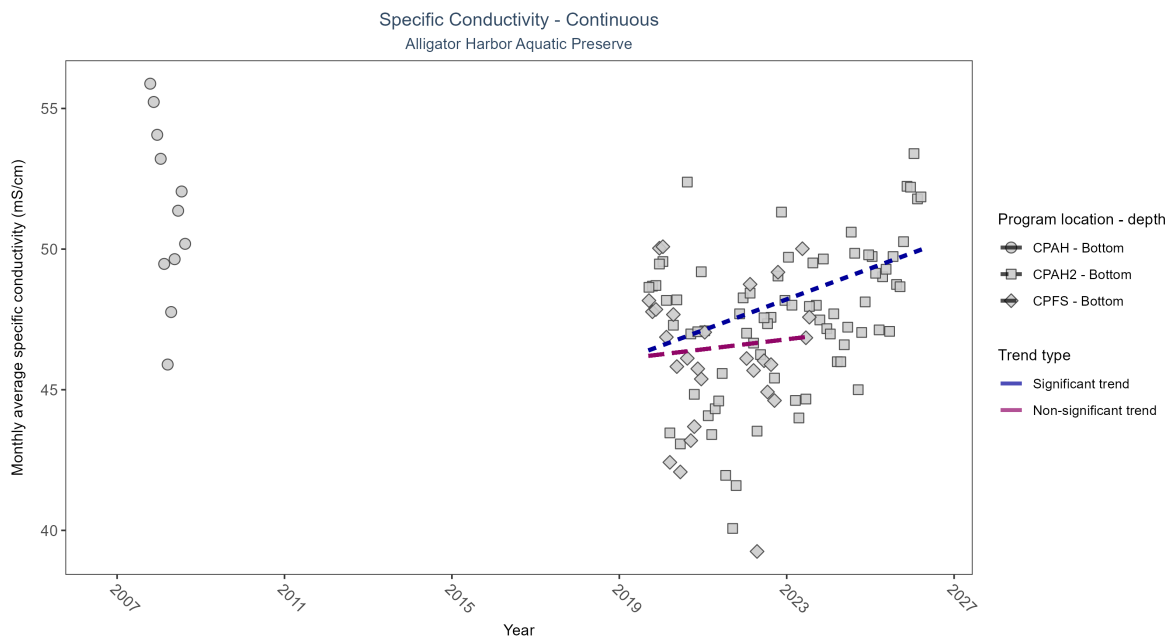


Figure 25: Scatter plot of monthly average specific conductivity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 12: At one program location, monthly average specific conductivity increased by 0.55 mS/cm per year. No detectable change in monthly average specific conductivity was observed at one location. There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
CPAH	Insufficient data	13034	2	2007 - 2008	51.89	-	-	-	-
CPAH2	Increasing trend	186148	8	2019 - 2026	47.93	0.31	46.02	0.55	0
CPFS	No detectable trend	60346	5	2019 - 2023	46.91	0	46.08	0.18	1

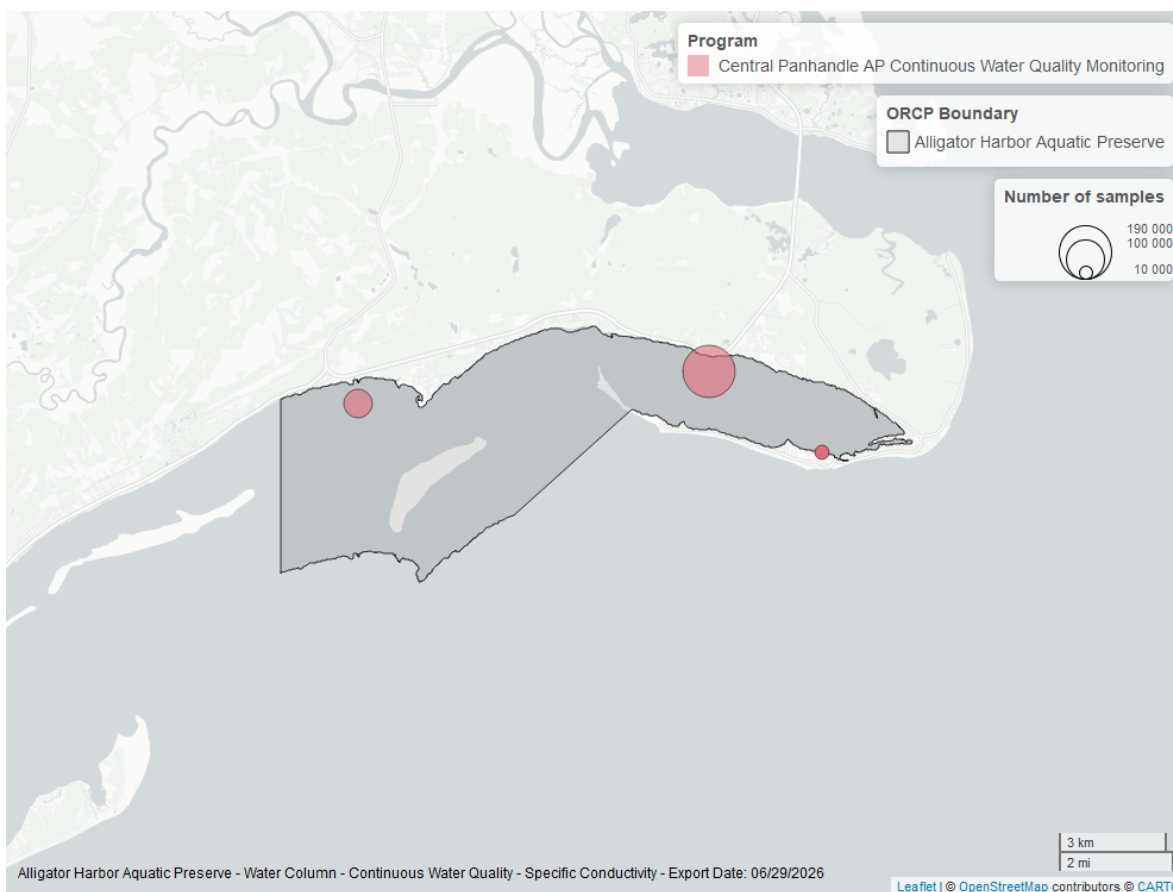


Figure 26: Map showing location of specific conductivity continuous water quality sampling locations within the boundaries of *Alligator Harbor Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Clarity

Turbidity - Continuous

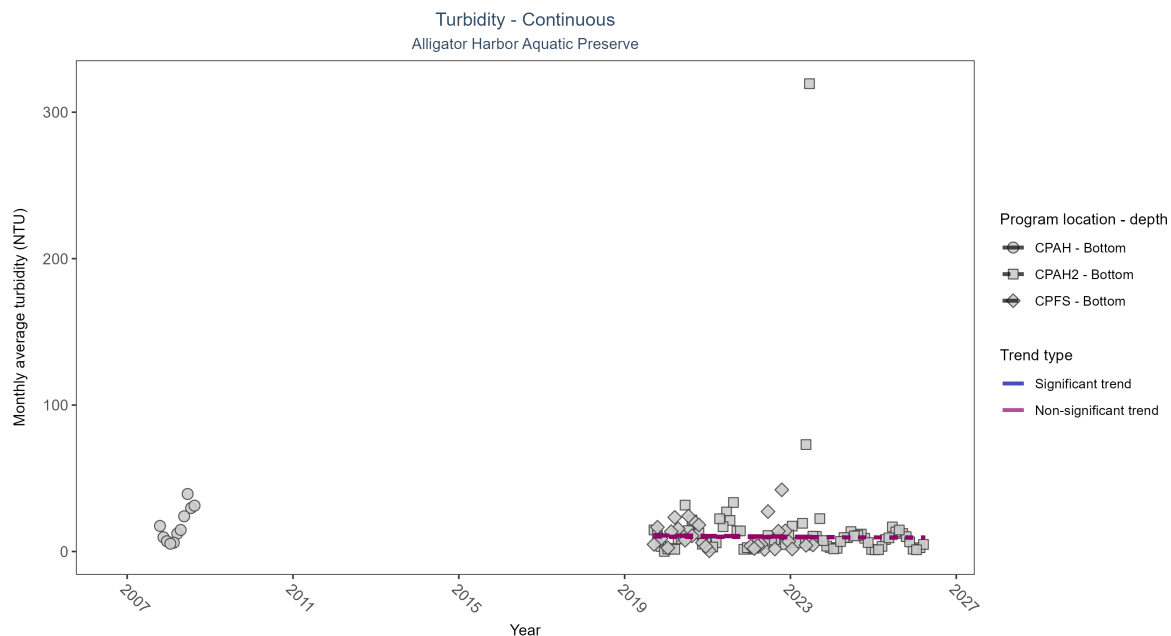


Figure 27: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 13: No detectable change in monthly average turbidity was observed at two locations. There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
CPAH	Insufficient data	12558	2	2007 - 2008	10	-	-	-	-
CPAH2	No detectable trend	191003	8	2019 - 2026	6	-0.06	10.19	-0.09	0.6
CPFS	No detectable trend	55980	5	2019 - 2023	3	-0.16	11.21	-0.26	1

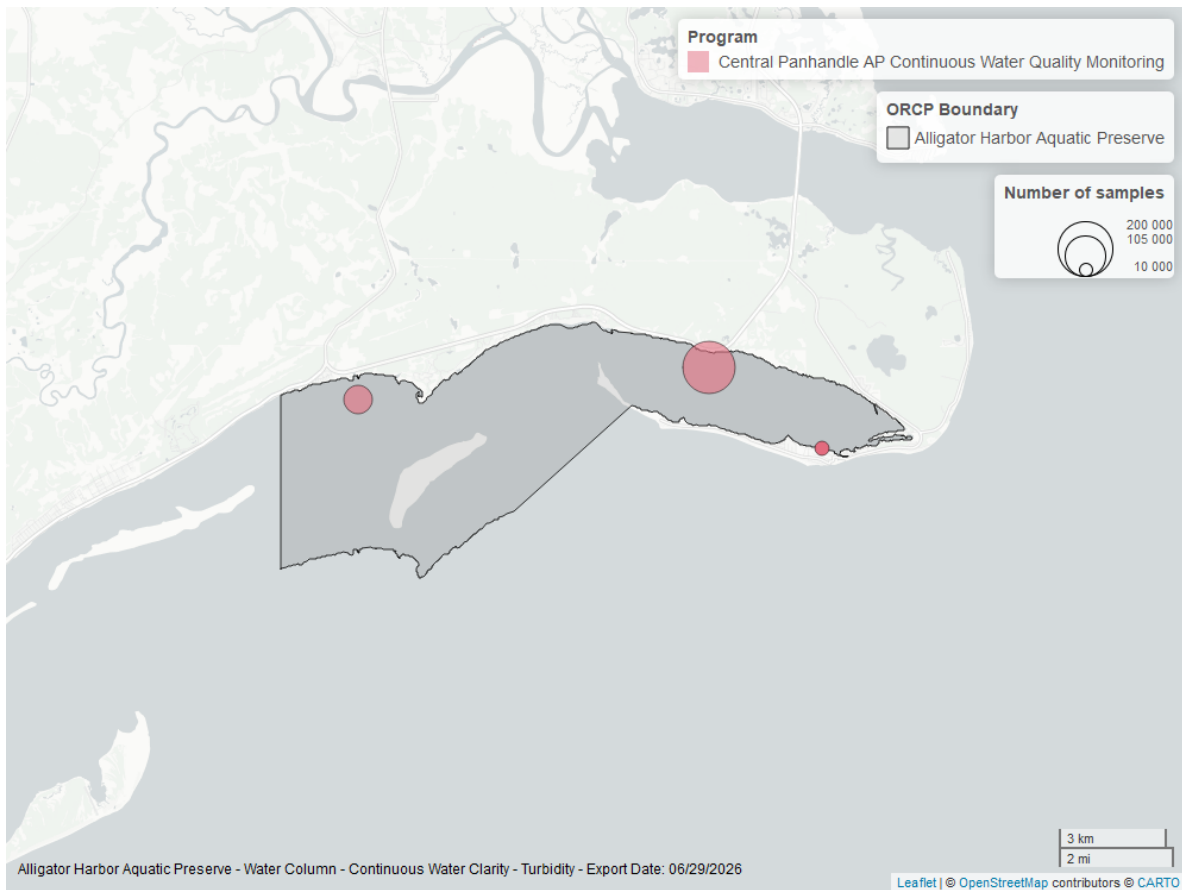


Figure 28: Map showing location of turbidity continuous water quality sampling locations within the boundaries of *Alligator Harbor Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

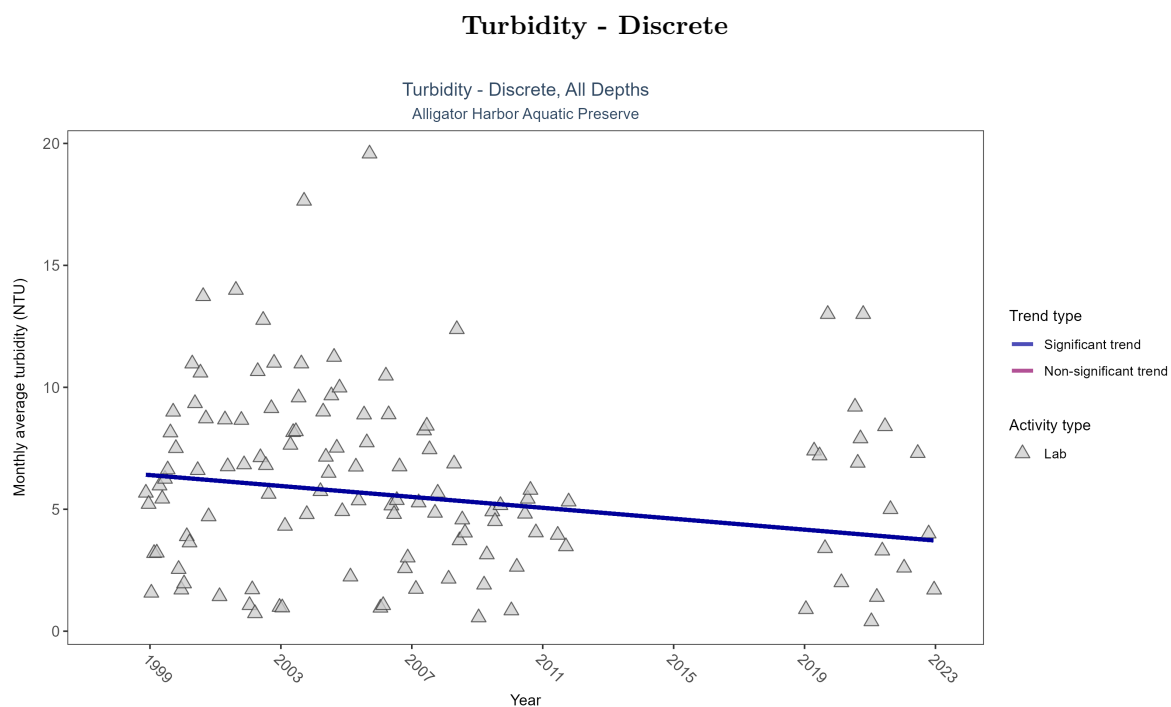


Figure 29: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 14: Monthly average turbidity decreased by 0.11 NTU per year, indicating an increase in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	3543	18	1998 - 2022	5.3	-0.24321	6.51312	-0.11185	0.0012

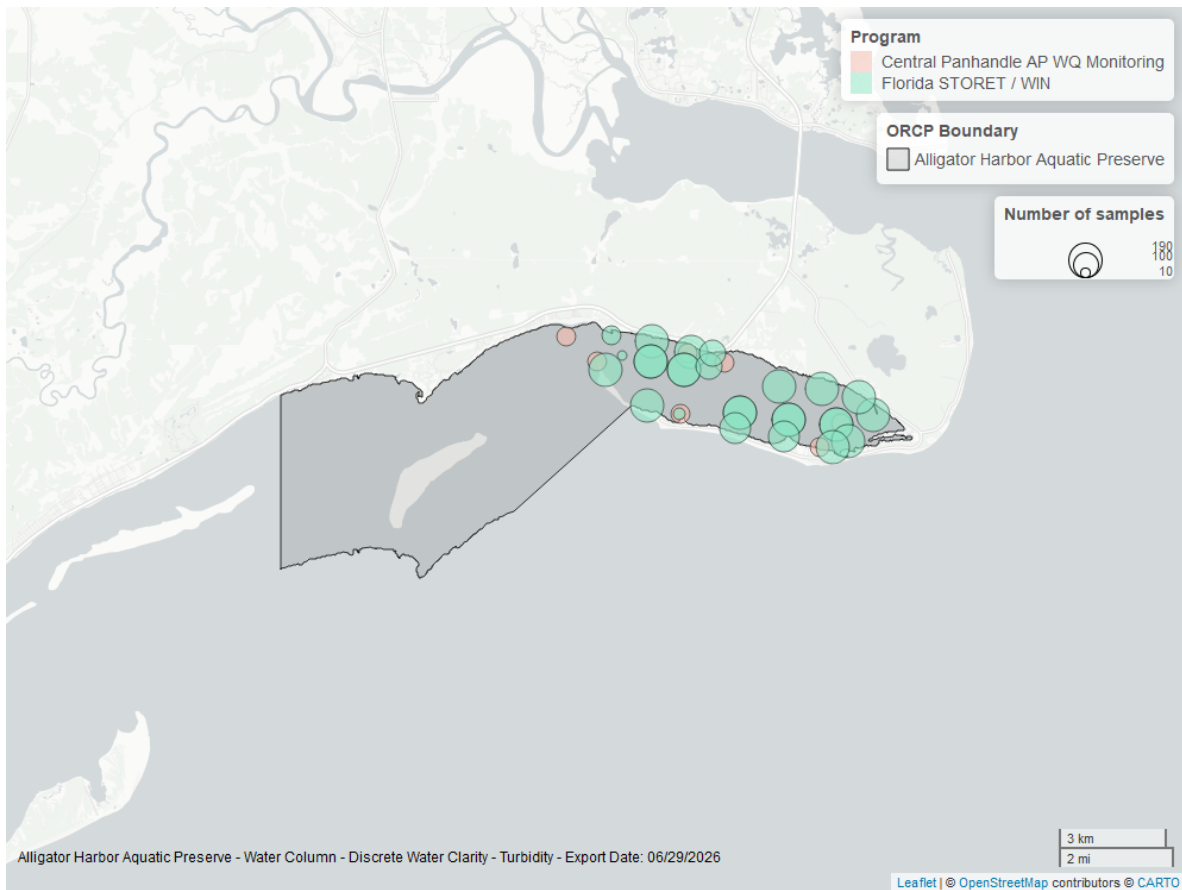


Figure 30: Map showing location of discrete water quality sampling locations within the boundaries of *Alligator Harbor Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

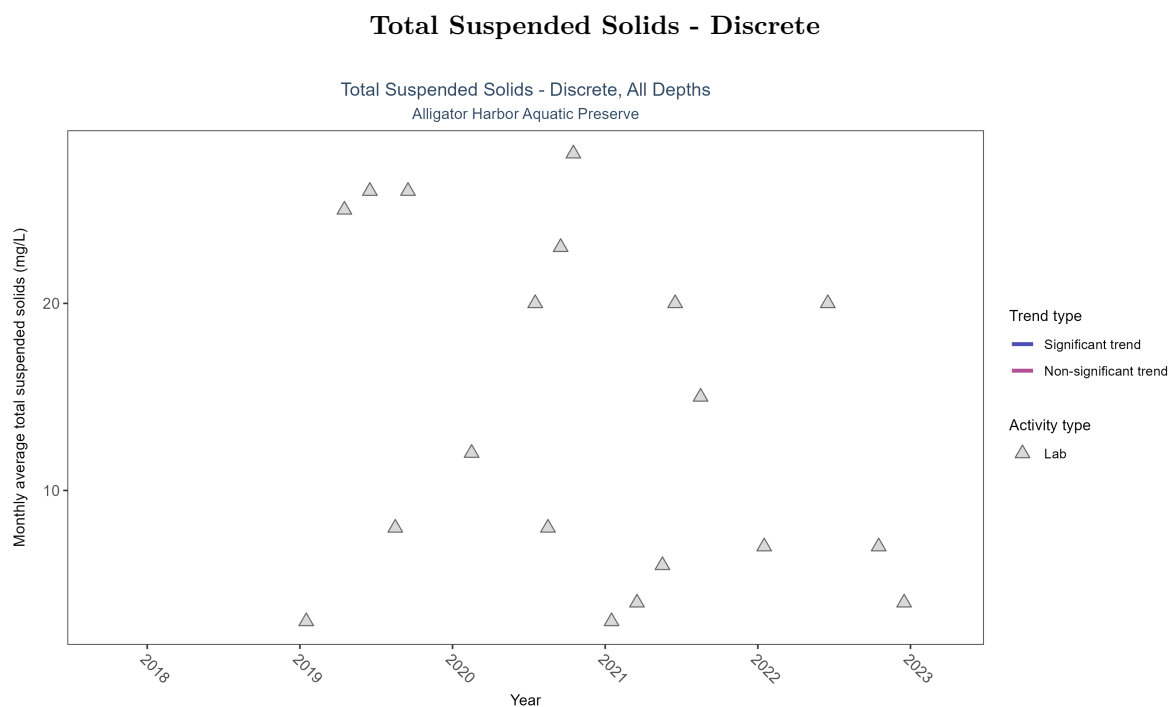


Figure 31: Scatter plot of monthly average total suspended solids (TSS) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only TSS values obtained from laboratory analyses (triangles) are included in the plot.

Table 15: There was insufficient data to fit a model for total suspended solids.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Insufficient data	19	4	2019 - 2022	12	-	-	-	-

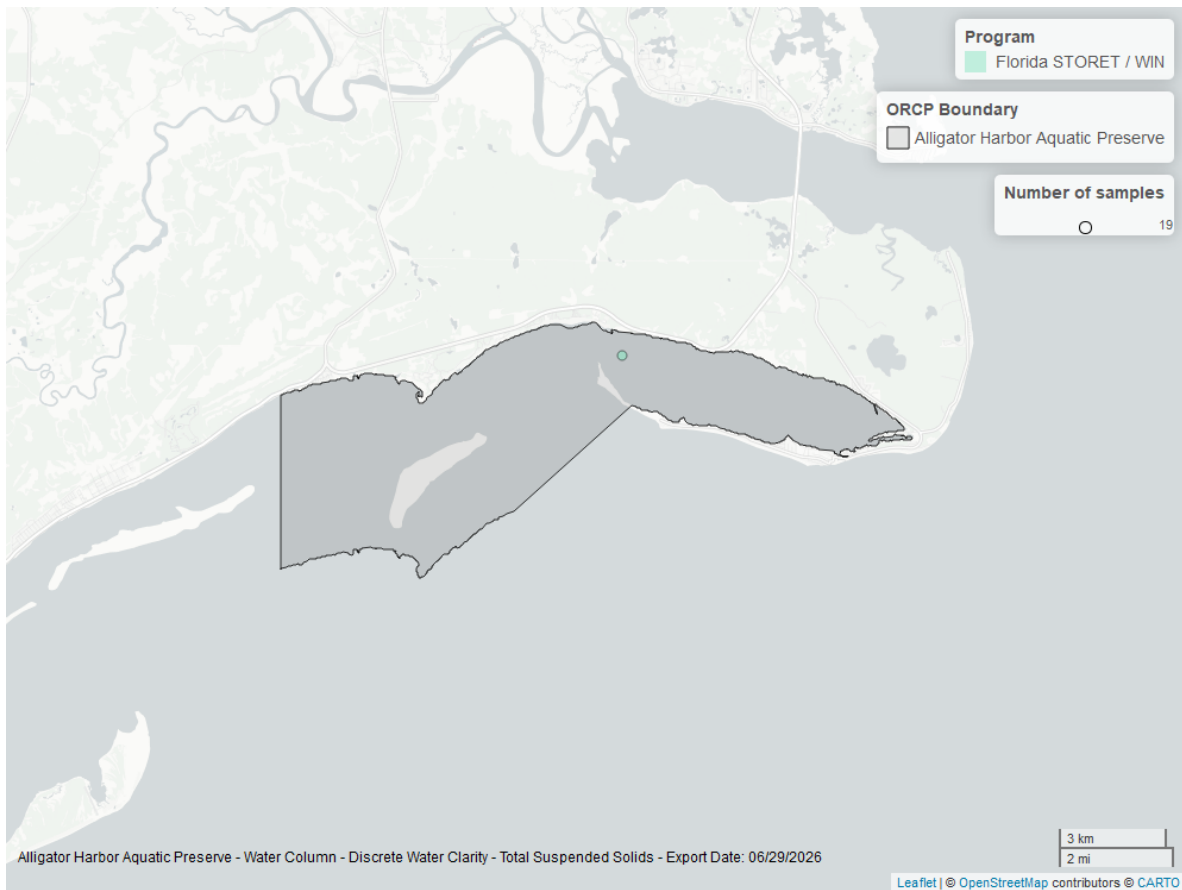


Figure 32: Map showing location of discrete water quality sampling locations within the boundaries of *Alligator Harbor Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Uncorrected for Pheophytin - Discrete

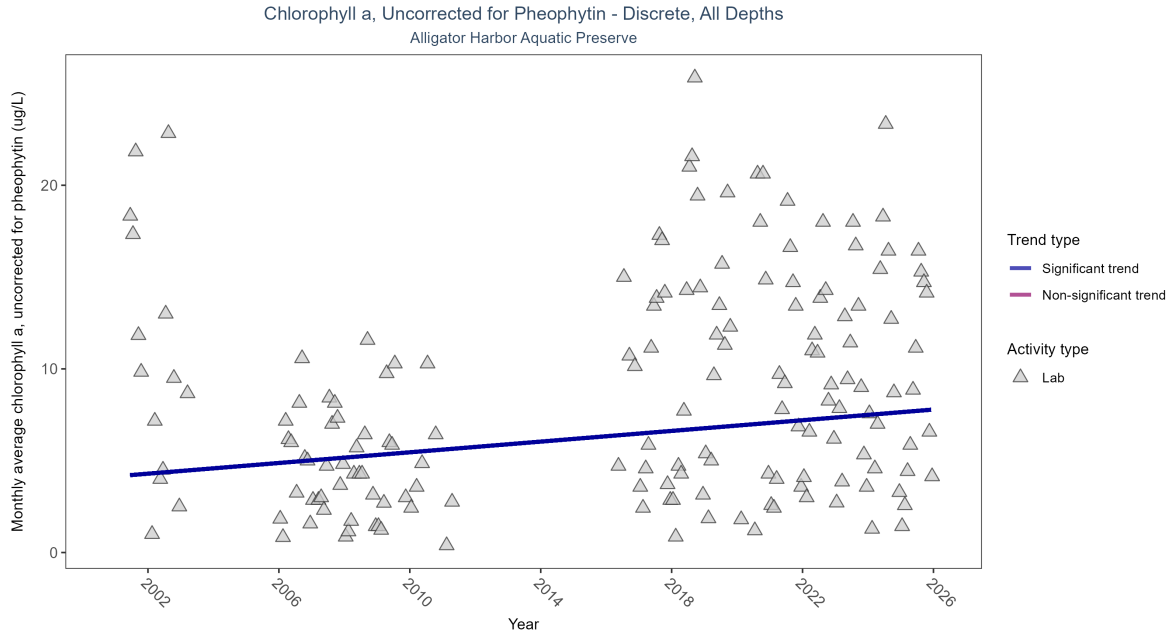


Figure 33: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 16: Monthly average chlorophyll a, uncorrected for pheophytin, increased by 0.15 ug/L per year, indicating a decrease in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	1213	19	2001 - 2025	6	0.27066	4.14792	0.14553	0

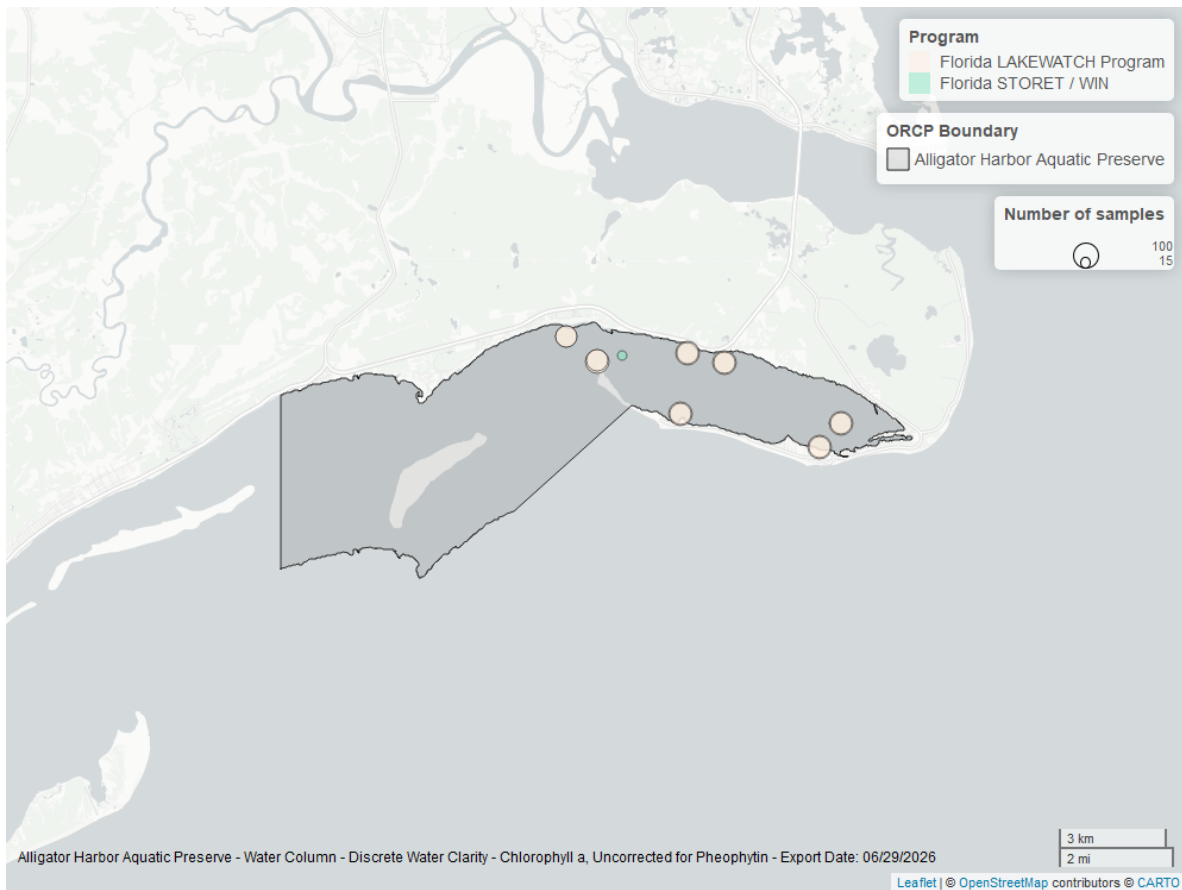


Figure 34: Map showing location of discrete water quality sampling locations within the boundaries of *Alligator Harbor Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Corrected for Pheophytin - Discrete

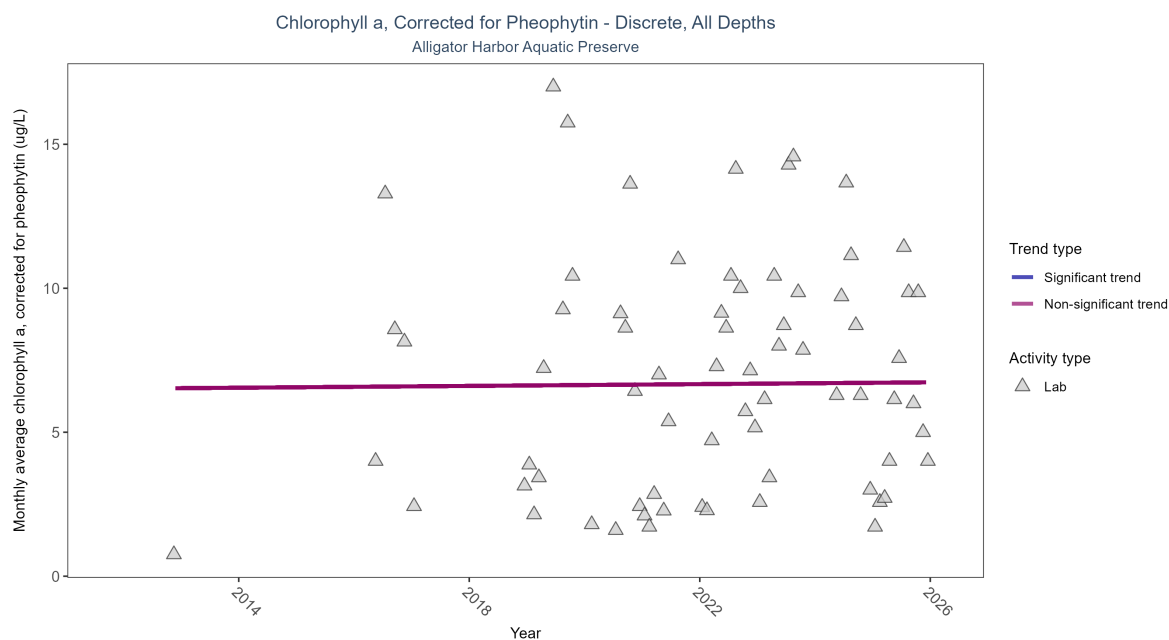


Figure 35: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 17: Chlorophyll a, corrected for pheophytin, showed no detectable trend between 2012 and 2025.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	471	11	2012 - 2025	6	0.01381	6.51434	0.01562	0.9116

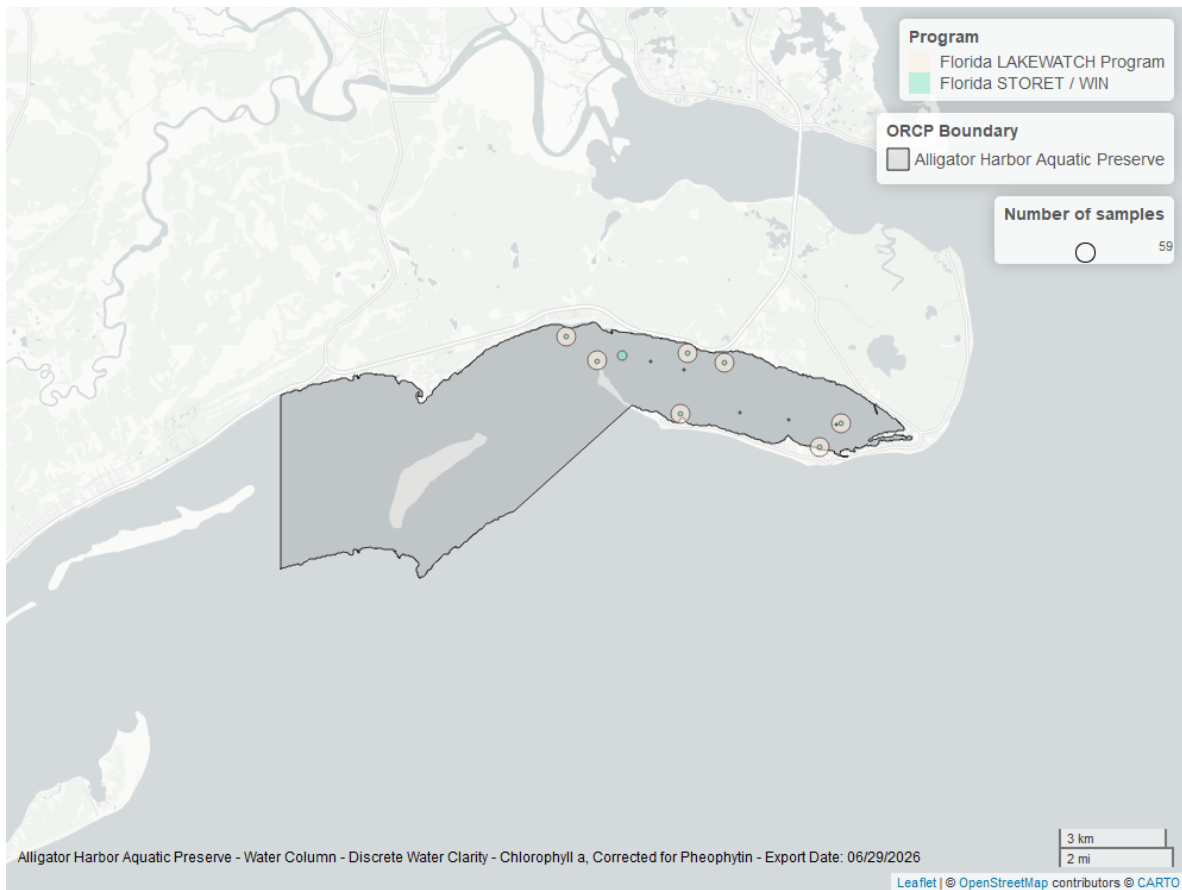


Figure 36: Map showing location of discrete water quality sampling locations within the boundaries of *Alligator Harbor Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

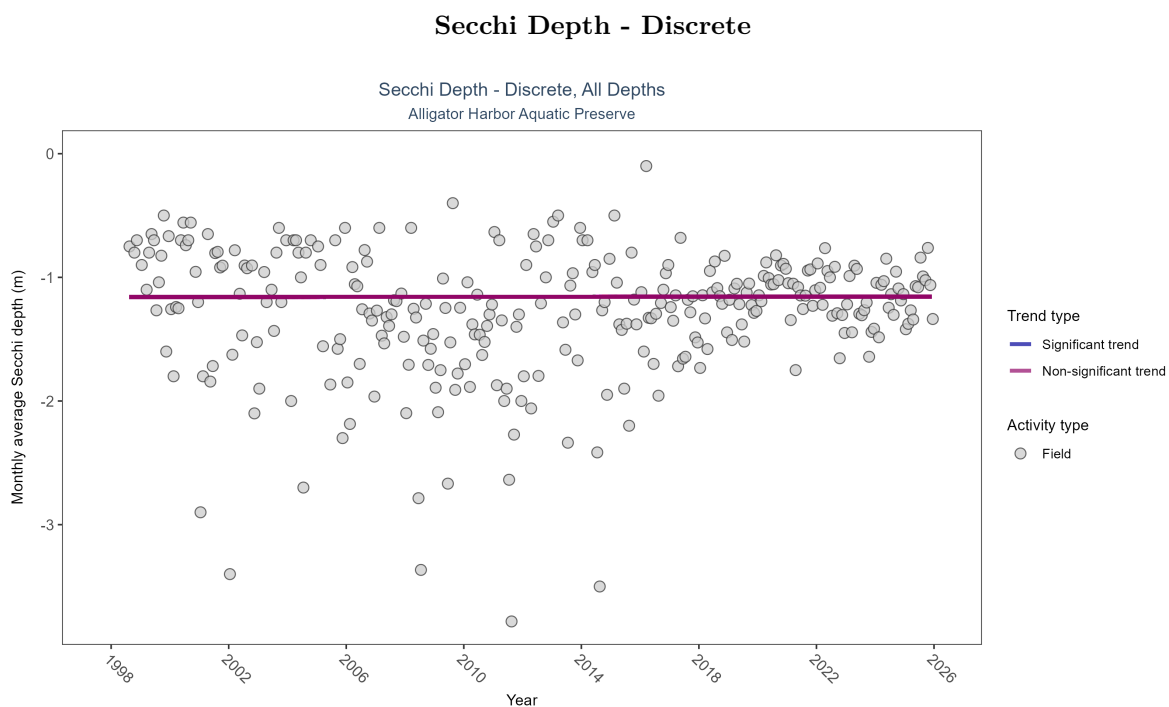


Figure 37: Scatter plot of monthly average Secchi depth over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Secchi depth is only measured in the field (circles).

Table 18: Secchi depth showed no detectable trend between 1998 and 2025.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	3154	28	1998 - 2025	-1.17	0.00942	-1.1589	0.00011	0.8716

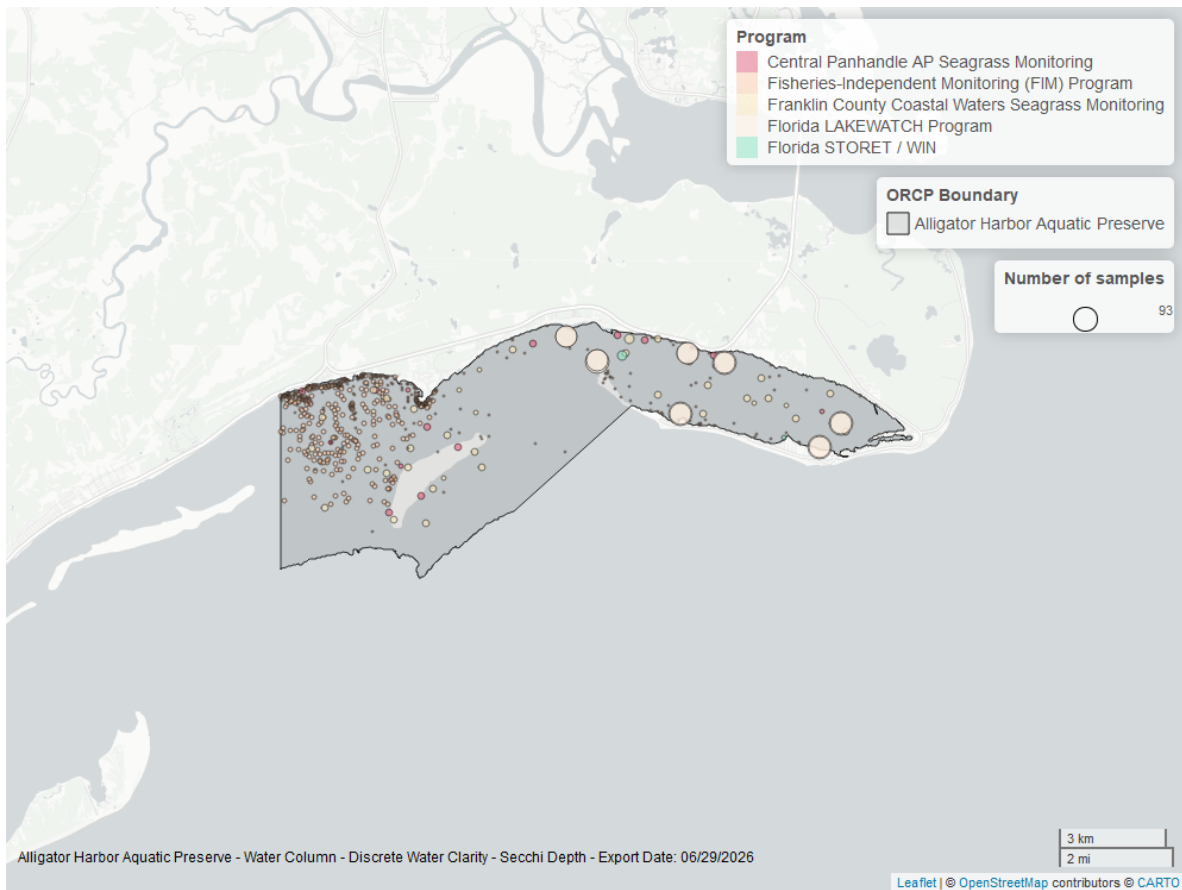


Figure 38: Map showing location of discrete water quality sampling locations within the boundaries of *Alligator Harbor Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Colored Dissolved Organic Matter - Discrete

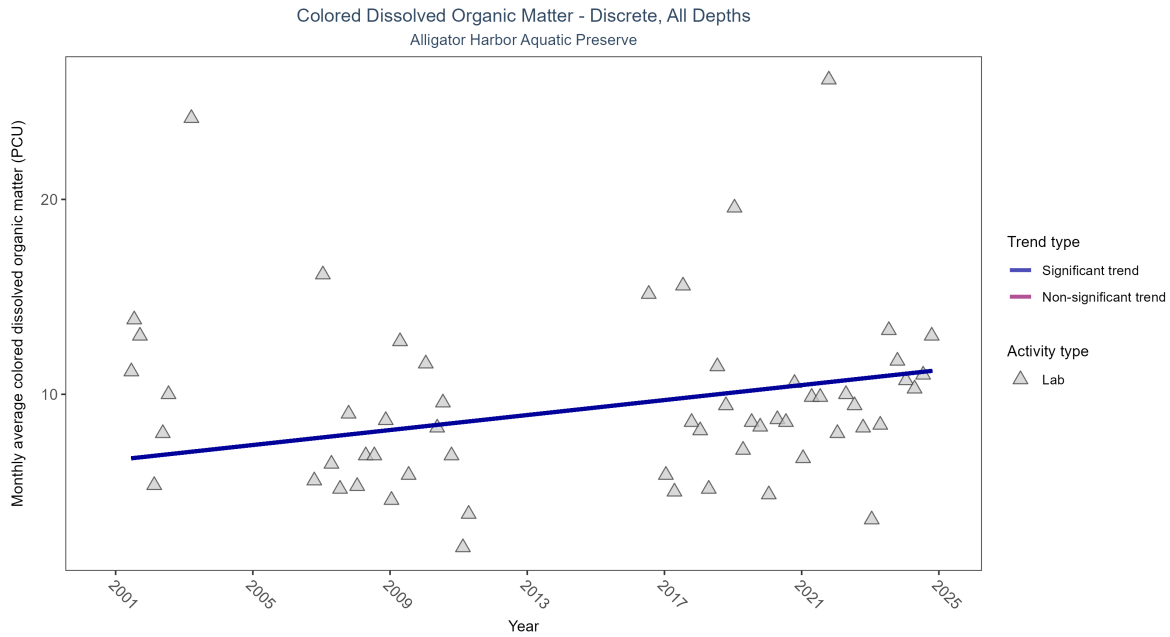


Figure 39: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 19: Monthly average colored dissolved organic matter increased by 0.19 PCU per year, indicating a decrease in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	423	18	2001 - 2024	8	0.23199	6.62963	0.19218	0.0315

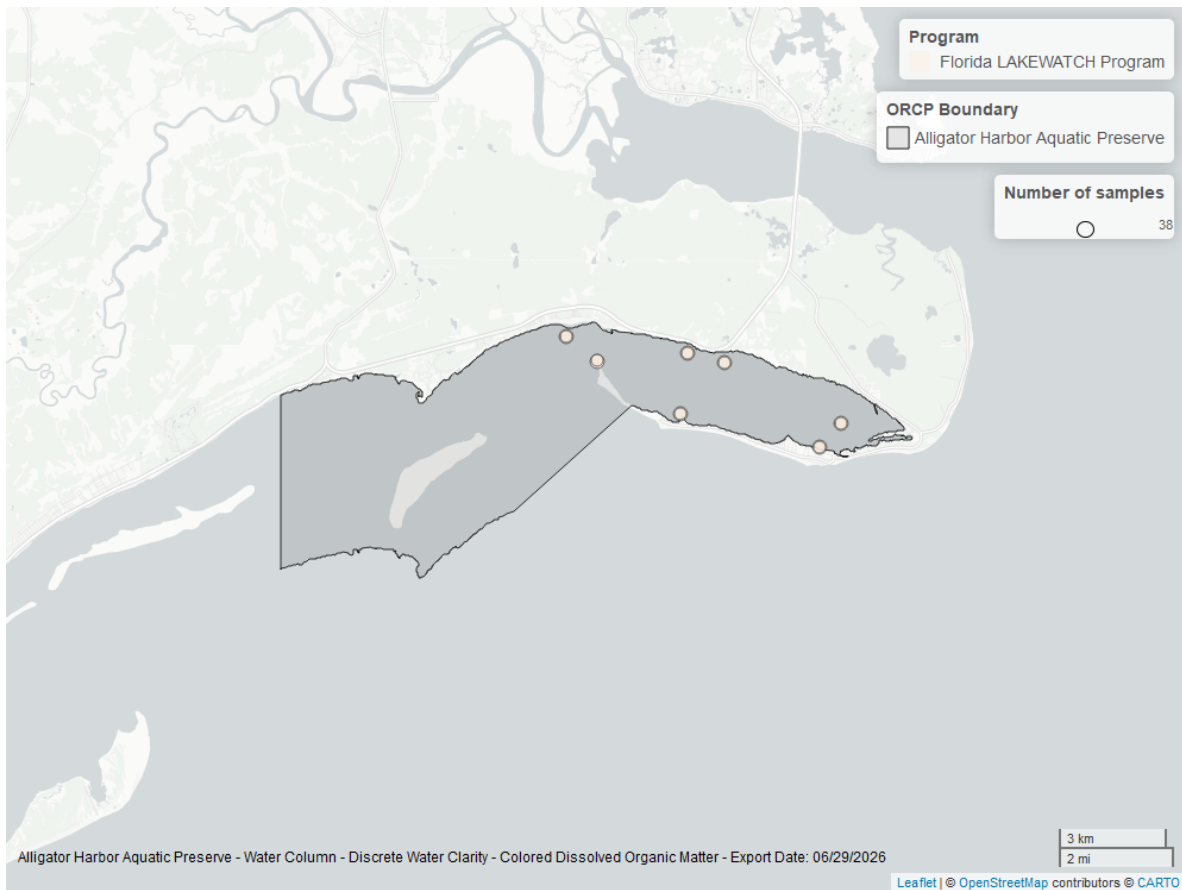


Figure 40: Map showing location of discrete water quality sampling locations within the boundaries of *Alligator Harbor Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.